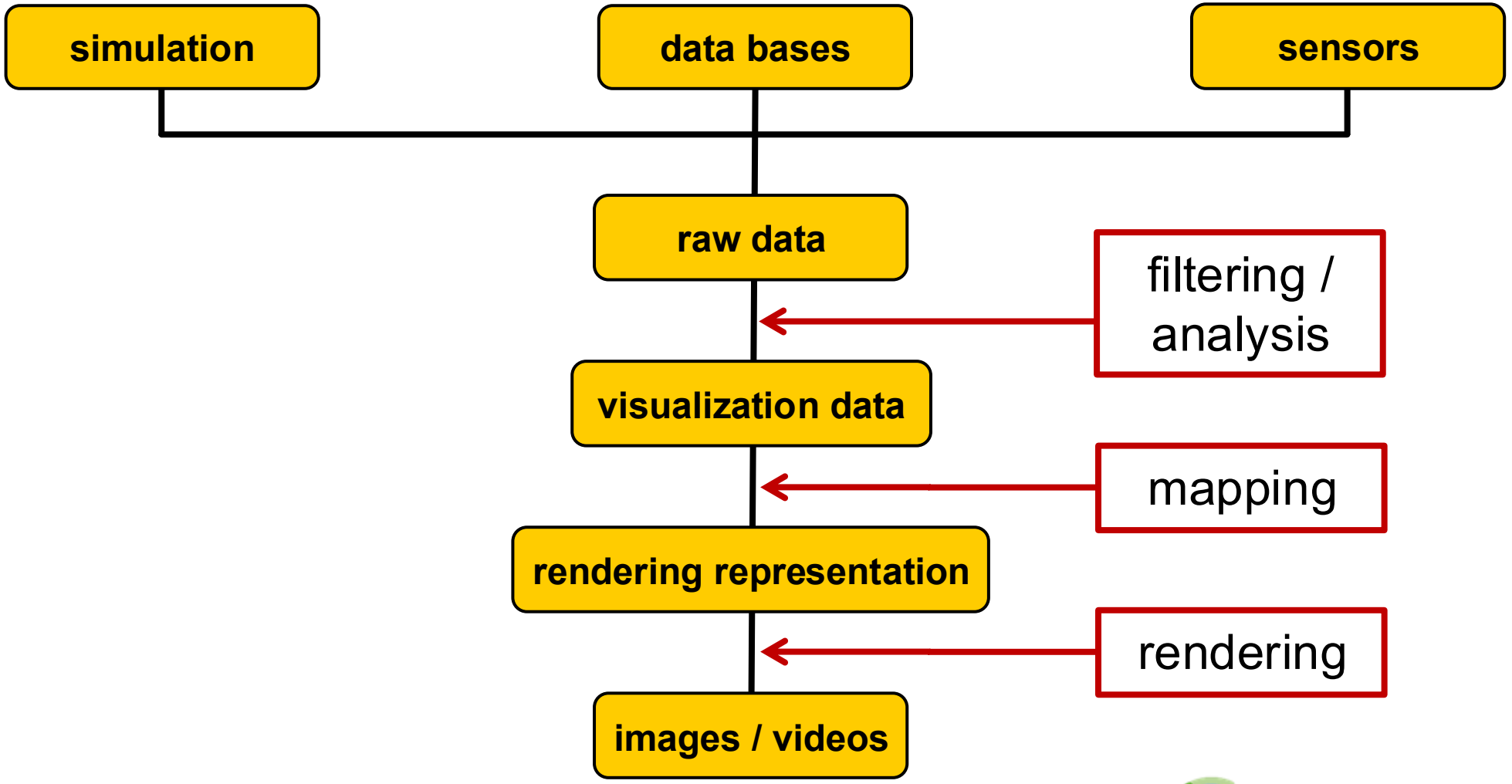
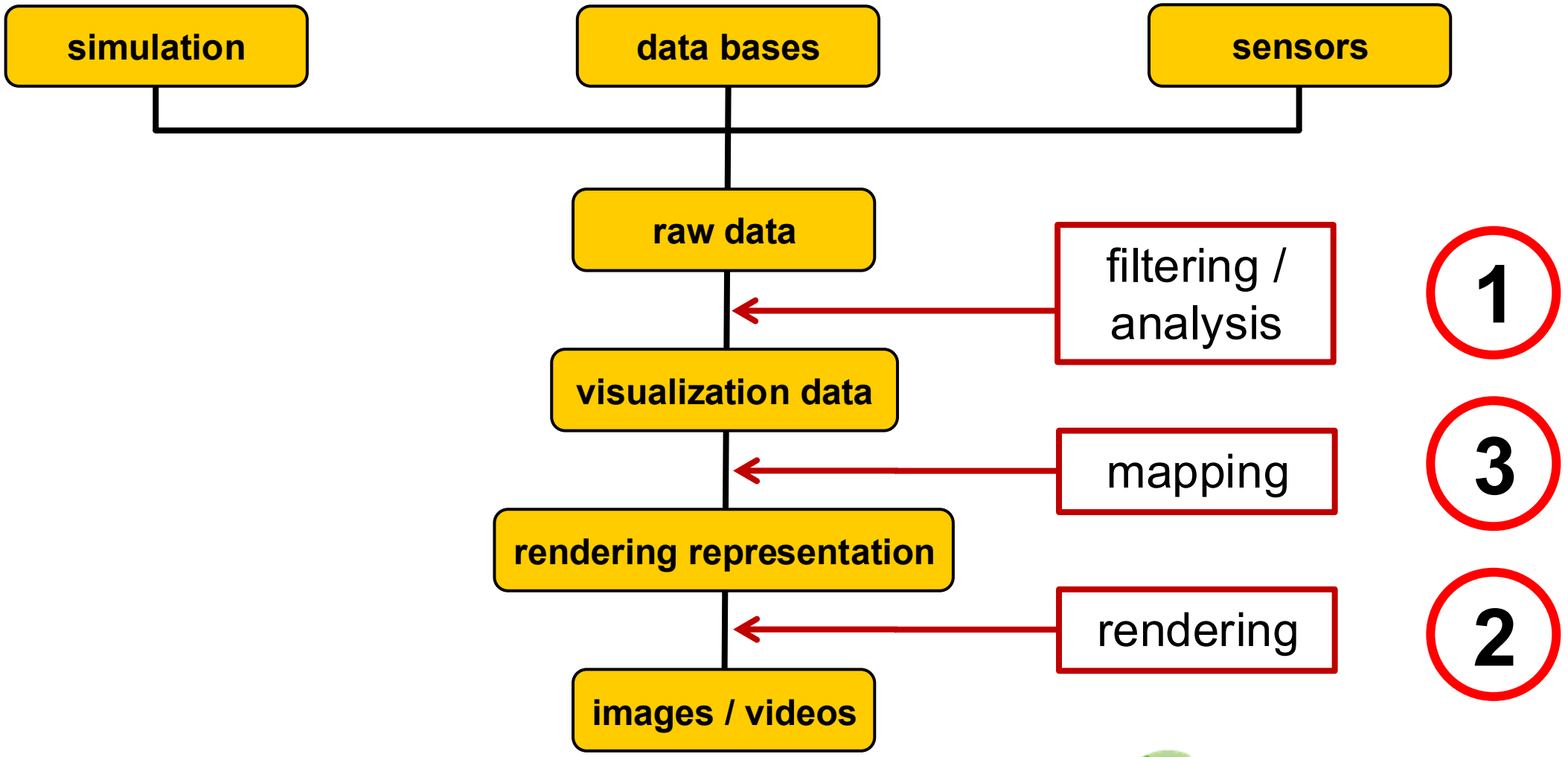
Data Analysis: Comparing Data

Visual Computing Institute
RWTH AACHEN UNIVERSITY

The Visualization Pipeline



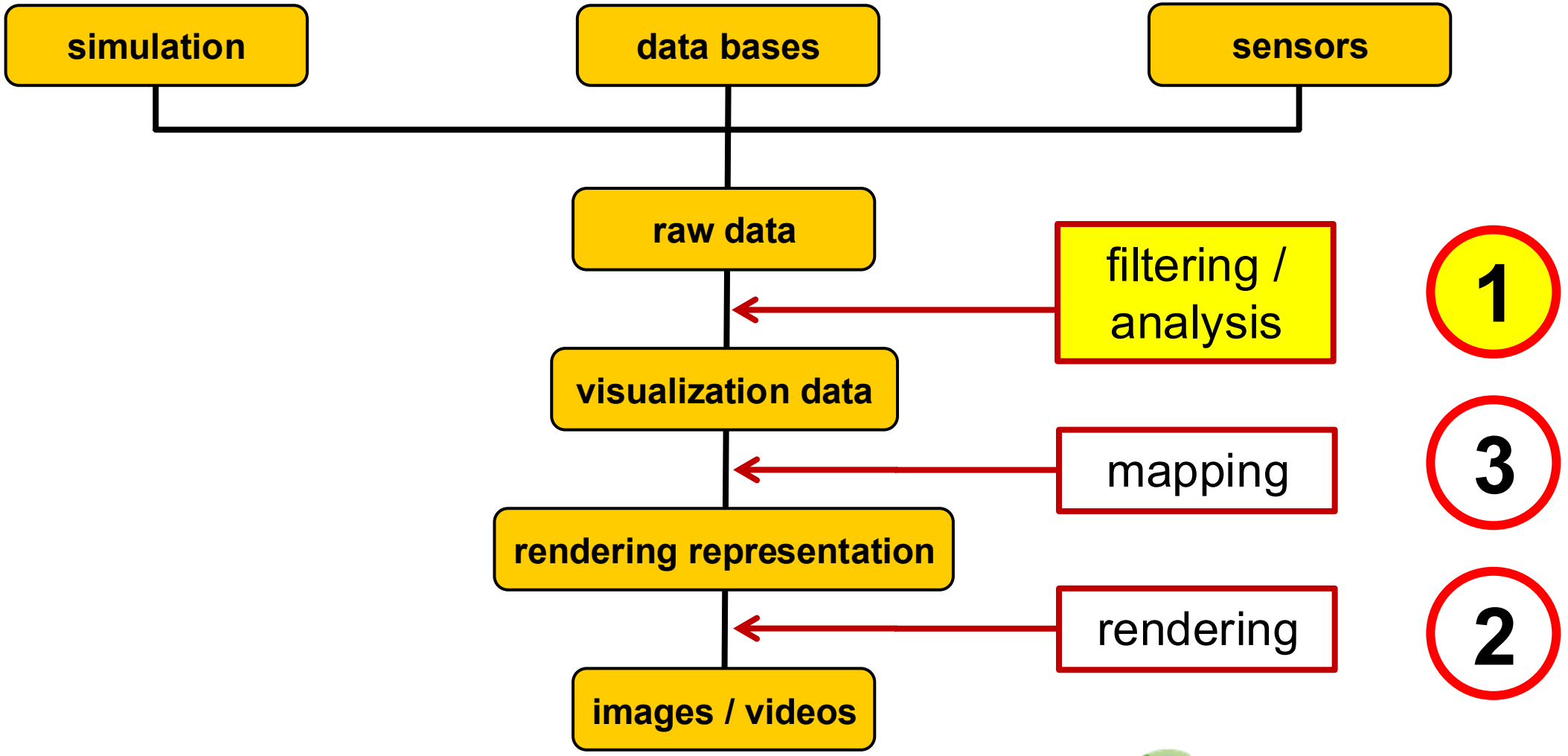
The Visualization Pipeline



Visual Computing Institute | Dr. Isaak Lim
Computer Graphics and Multimedia
Data Analysis and Visualization

Visual Computing Institute
RWTH AACHEN UNIVERSITY

The Visualization Pipeline



Motivation and Mission

Data Analysis: finding structure in data

- are there any similarities in my dataset? → *comparison*
- are there patterns and structures → *dimensionality reduction*
- can we identify groups of similar data items → *clustering*

Data Analysis: Outline

- Types of Data
- Comparing Data
- Dimensionality Reduction
- Clustering

Data Analysis: Outline

- **Types of Data**
- Comparing Data
- Dimensionality Reduction
- Clustering

Types of Data

- most facts / relations / models / objects …
  can mathematically be conceptualized as functions:
- $f\colon \Omega \subseteq \mathbb{R}^d \to \Gamma \subseteq \mathbb{R}^k$

- data is typically a **set** of *discrete* samples / measurements:
- $f(\omega_1),\ \dots,\ f(\omega_n) \in \mathbb{R}^k$

- or a **set of sets** of samples:
- $f_1(\omega_{ij}), \dots, f_n(\omega_{ij})$

Types of Data

- most facts / relations / models / objects …
  can mathematically be conceptualized as functions:
- $f\colon \Omega \subseteq \mathbb{R}^d \to \Gamma \subseteq \mathbb{R}^k$

Sometimes Ω or Γ are not Euclidean but, e.g., hyperspheres or other

- data is typically a **set** of *discrete* samples / measurements:
- $f(\omega_1), \ \dots, \ f(\omega_n) \in \mathbb{R}^k$

- or a **set of sets** of samples:
- $f_1(\omega_{ij}), \dots, f_n(\omega_{ij})$

Types of Data

- most facts / relations / models / objects …
  can mathematically be conceptualized as functions:
- $f\colon \Omega \subseteq \mathbb{R}^d \to \Gamma \subseteq \mathbb{R}^k$

Sometimes Ω or Γ are not Euclidean but, e.g., hyperspheres or other

- data is typically a **set** of *discrete* samples / measurements:
- $f(\omega_1), \ \dots, \ f(\omega_n) \in \mathbb{R}^k$

- or a **set of sets** of samples:
- $f_1(\omega_{ij}), \dots, f_n(\omega_{ij})$

Sampling pattern ω_{ij} can be individual or the same (e.g., pixel grid)

Types of Data

- data types can be classified by the dimension of their domain:
  - d = 1 : univariate („curve“, „time series“, …)
  - d = 2 : bivariate („surface“, …)
  - d = 3 : trivariate („volume“, …)
  - …

- or the dimension of their range:
  - k = 1 : scalar
  - k = 2 : planar
  - k = 3 : spatial
  - …

$$f\colon \Omega \subseteq \mathbb{R}^d \to \Gamma \subseteq \mathbb{R}^k$$

Examples of Types of Data

- *sound*

Examples of Types of Data

- *sound* = pressure over time (d=1, k = 1)
- *terrain*

Examples of Types of Data

- *sound* = pressure over time (d=1, k = 1)
- *terrain* = height over sea level (d=2, k = 1)
- *image*

Examples of Types of Data

- *sound* = pressure over time (d=1, k = 1)
- *terrain* = height over sea level (d=2, k = 1)
- *image* = RGB over 2D grid (d=2, k = 3)
- *3D shape*

Examples of Types of Data

- *sound* = pressure over time (d=1, k = 1)
- *terrain* = height over sea level (d=2, k = 1)
- *image* = RGB over 2D grid (d=2, k = 3)
- *3D shape* = XYZ over UV-domain (d=2, k=3)
- *3D shape*

Examples of Types of Data

- *sound* = pressure over time (d=1, k = 1)
- *terrain* = height over sea level (d=2, k = 1)
- *image* = RGB over 2D grid (d=2, k = 3)
- *3D shape* = XYZ over UV-domain (d=2, k=3)
- *3D shape* = distance over XYZ-domain (d=3, k=1)
- *flow in space*

Examples of Types of Data

- *sound* = pressure over time (d=1, k = 1)
- *terrain* = height over sea level (d=2, k = 1)
- *image* = RGB over 2D grid (d=2, k = 3)
- *3D shape* = XYZ over UV-domain (d=2, k=3)
- *3D shape* = distance over XYZ-domain (d=3, k=1)
- *flow in space* = XYZ over XYZ (d=3, k=3)

Data Analysis: Outline

- Types of Data
- **Comparing Data**
  - **feature spaces**
  - similarity and dissimilarity measures
  - comparing histograms and distributions
- Dimensionality Reduction
- Clustering

“Simple Data”

- data samples $s_i = f(\omega_i) \in \mathbb{R}^k$
  - comparison by some metric in $\mathbb{R}^k$ (e.g Euclidian distance)
  - may first normalize properties by affine maps $A(s) = m\,s + t$
  - embed samples into feature space $l_i = M\,s_i$

    (M could be a neural network)

Similarity Model

Γ : universe of elements in dataset
$\mathbb{F}$: feature space

- similarity model $S: \Gamma \times \Gamma \rightarrow \mathbb{R}$

Similarity Model

Γ : universe of elements in dataset
$\mathbb{F}$: feature space

- similarity model $S: \Gamma \times \Gamma \rightarrow \mathbb{R}$
  - $p, q \in \Gamma$

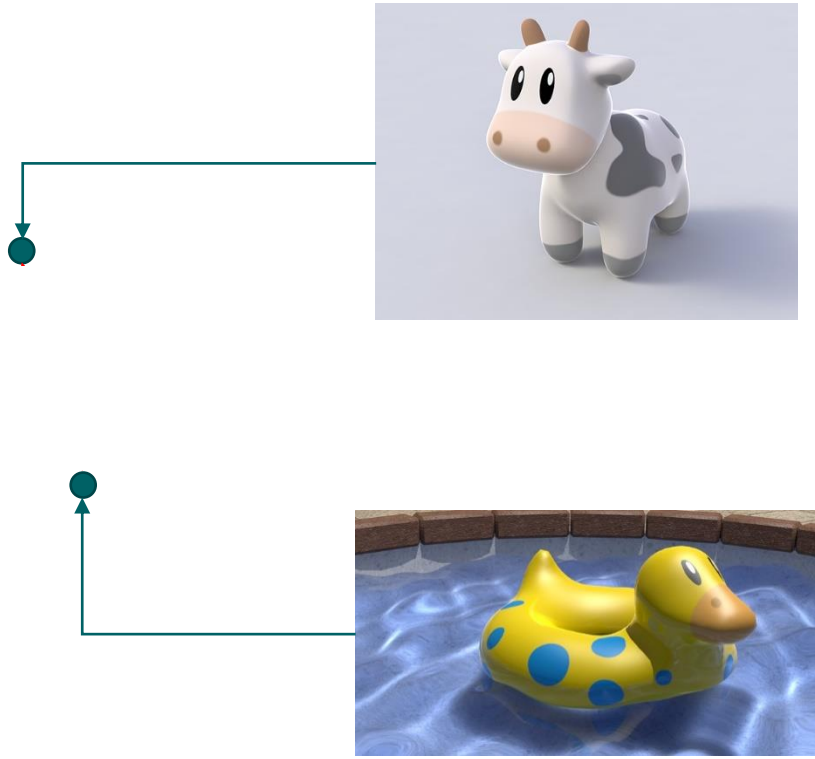




Similarity Model

Γ : universe of elements in dataset
$\mathbb{F}$: feature space

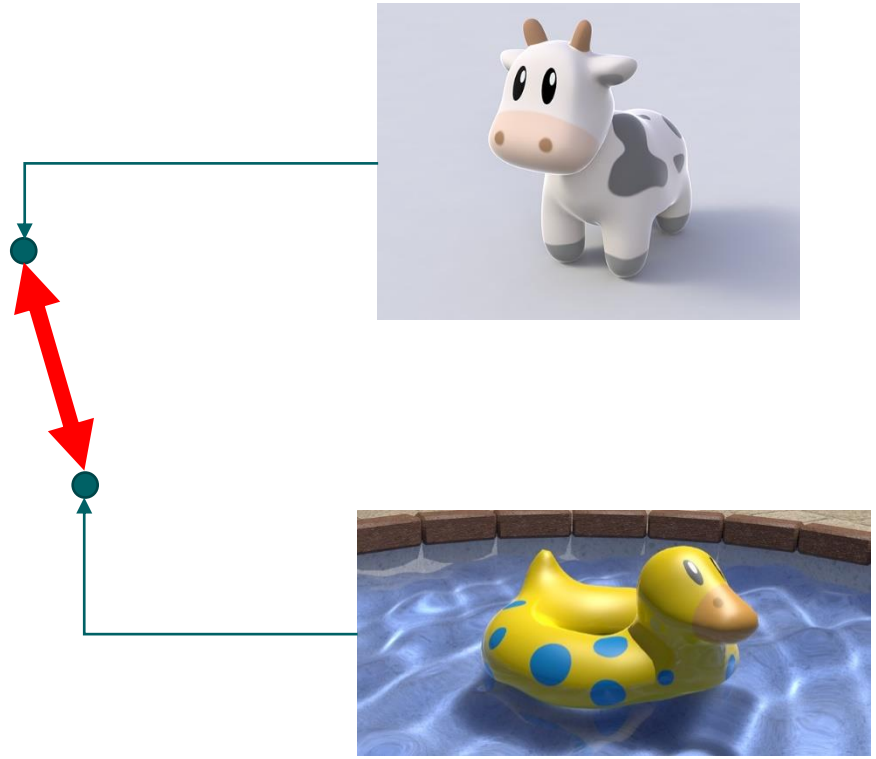
- similarity model $S: \Gamma \times \Gamma \rightarrow \mathbb{R}$
  - $p, q \in \Gamma$
  - feature extraction: $f: \Gamma \rightarrow \mathbb{F}$



Similarity Model

Γ : universe of elements in dataset
$\mathbb{F}$: feature space

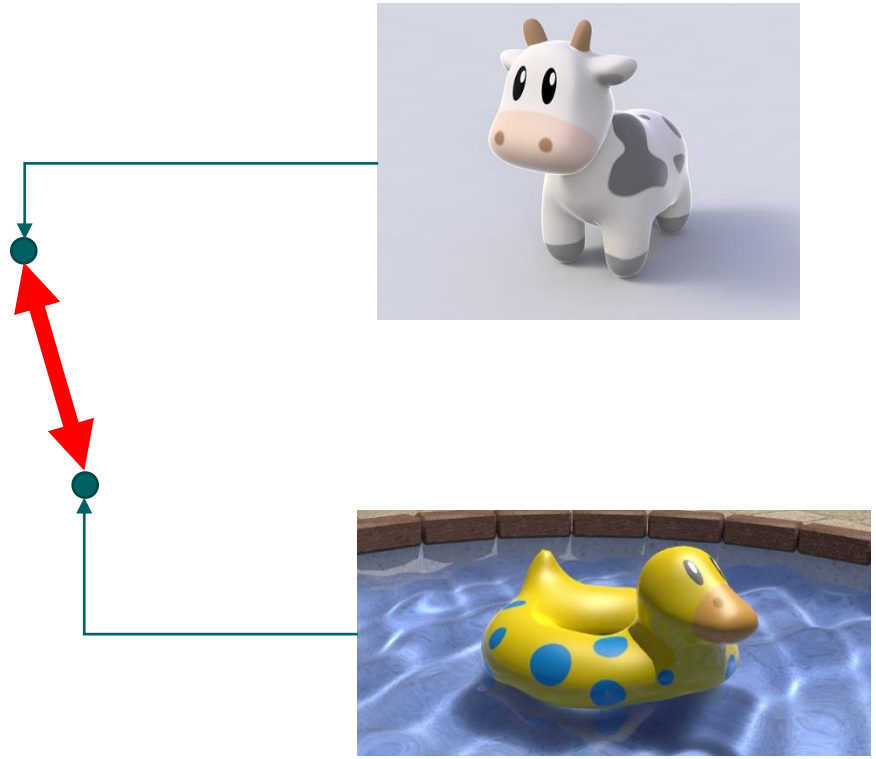
- similarity model $S: \Gamma \times \Gamma \rightarrow \mathbb{R}$
  - $p, q \in \Gamma$
  - feature extraction: $f: \Gamma \rightarrow \mathbb{F}$
  - similarity measure: $sim: \mathbb{F} \times \mathbb{F} \rightarrow \mathbb{R}_{\geq 0}$



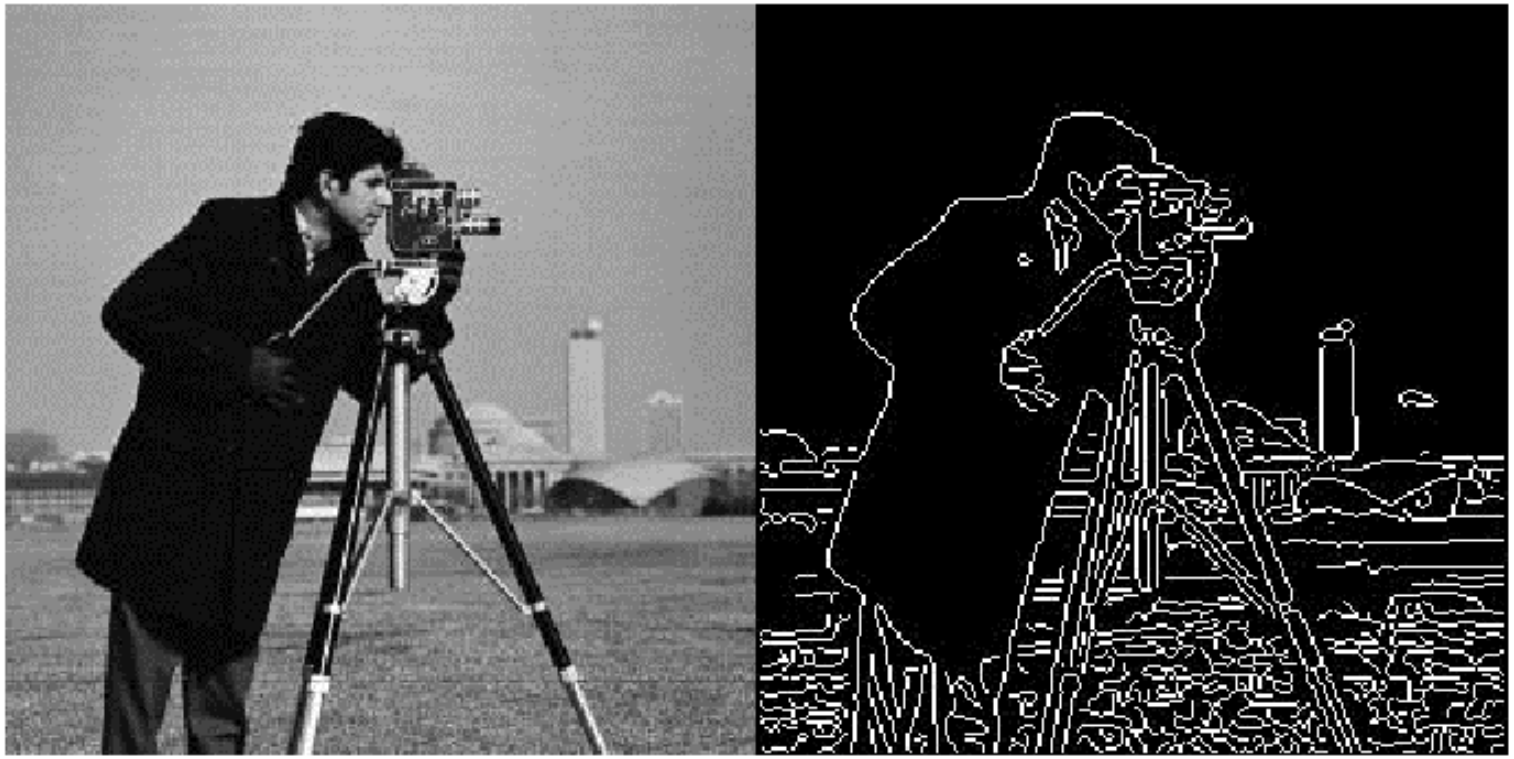
Similarity Model

Γ : universe of elements in dataset
$\mathbb{F}$: feature space

- similarity model $S: \Gamma \times \Gamma \rightarrow \mathbb{R}$
  - $p, q \in \Gamma$
  - feature extraction: $f: \Gamma \rightarrow \mathbb{F}$
  - similarity measure: $sim: \mathbb{F} \times \mathbb{F} \rightarrow \mathbb{R}_{\geq 0}$
  - $S(p,q) = sim\big(f(p), f(q)\big)$



Feature Extraction: Images … Edges



Canny Edge Detector

Separate relevant from irrelevant information.

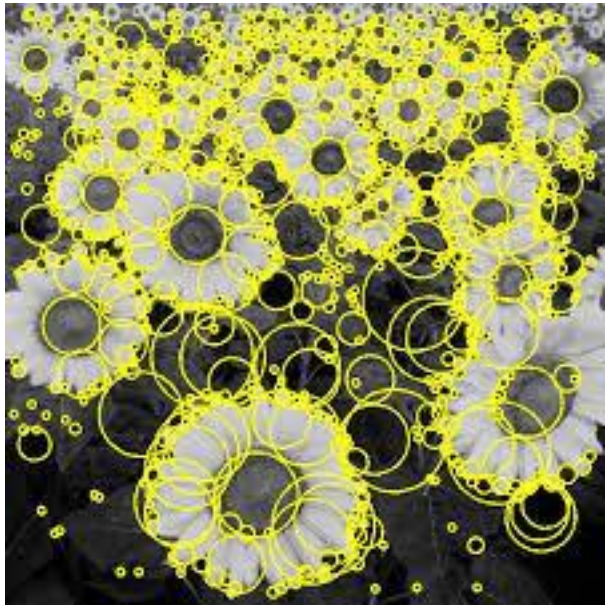
Feature Extraction: Images … Corners



Harris Corner Detector

Feature Extraction: Images … Interest points



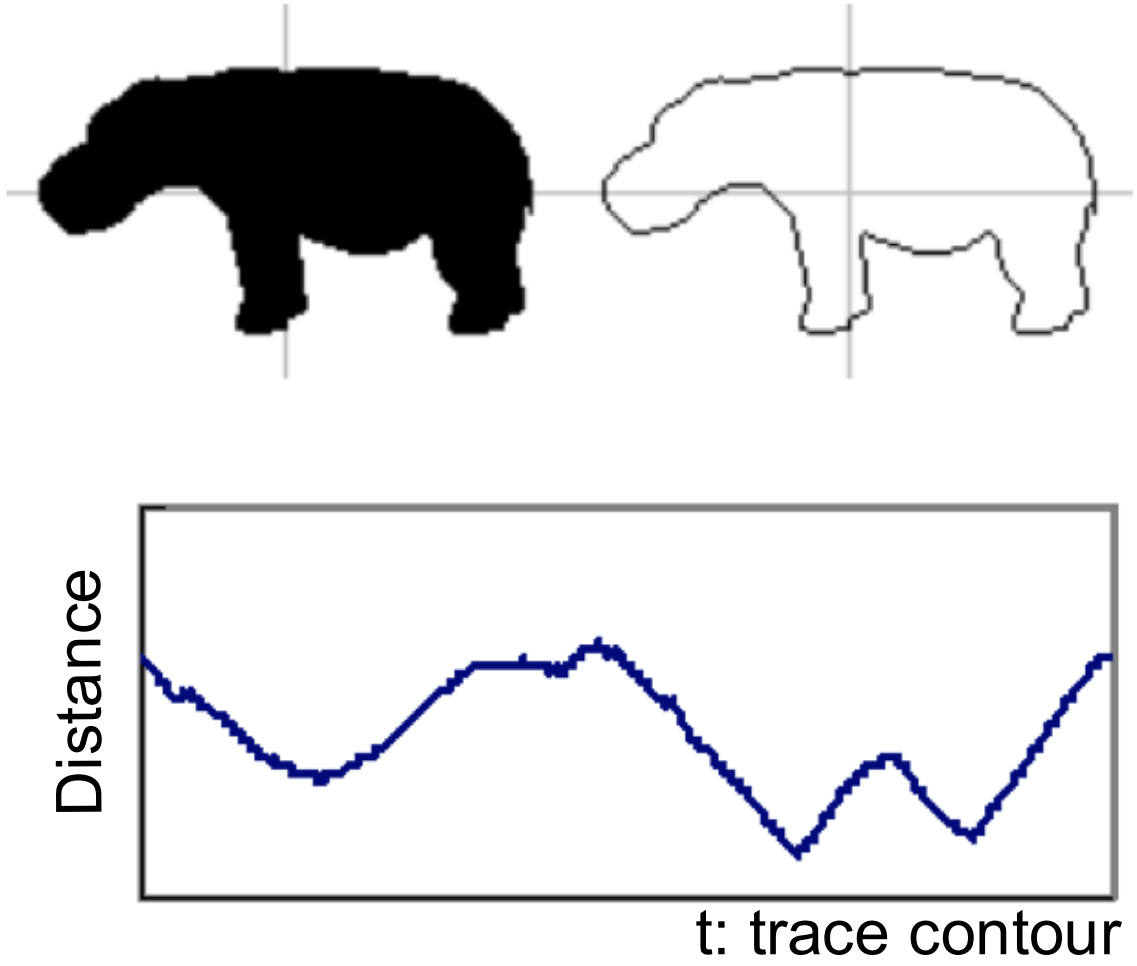


SIFT Feature Detector

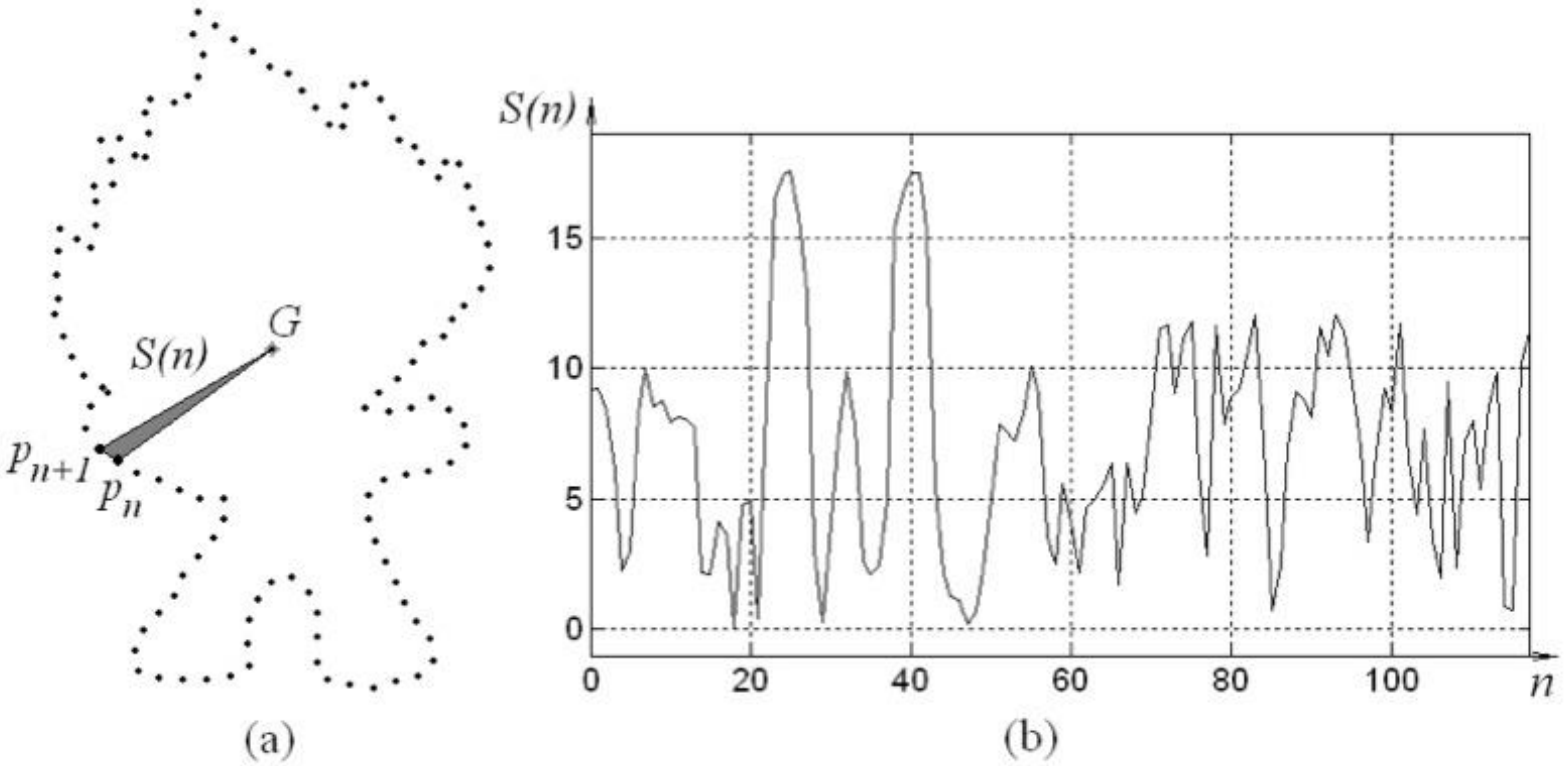
Visual Computing Institute | Dr. Isaak Lim
Computer Graphics and Multimedia
Data Analysis and Visualization

Visual Computing Institute | RWTH AACHEN UNIVERSITY

Feature Extraction: Shapes … Feature Signature Functions



Feature Extraction: Shapes … Feature Signature Functions

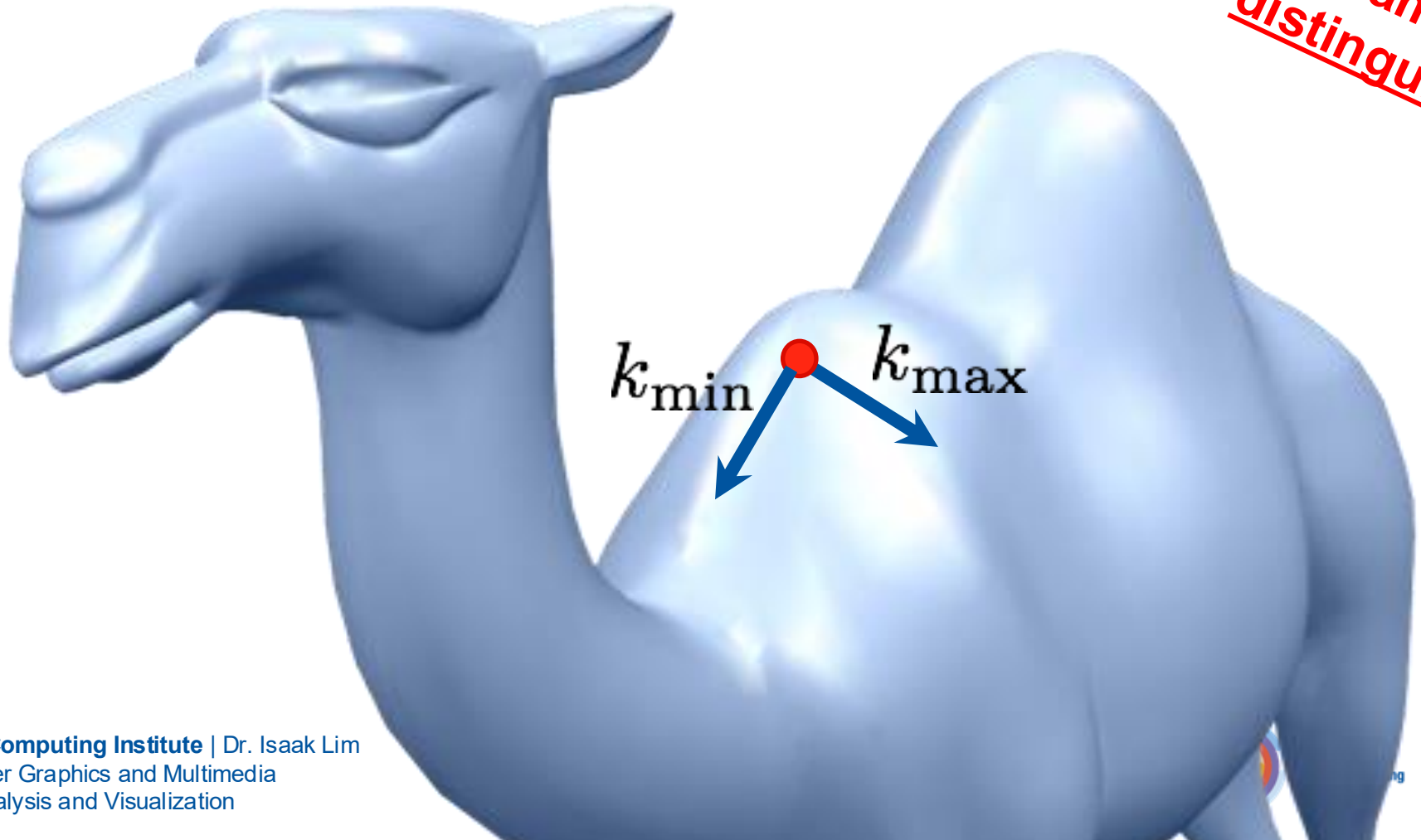


Make samples compatible for comparison.

Feature Extraction: Curvature

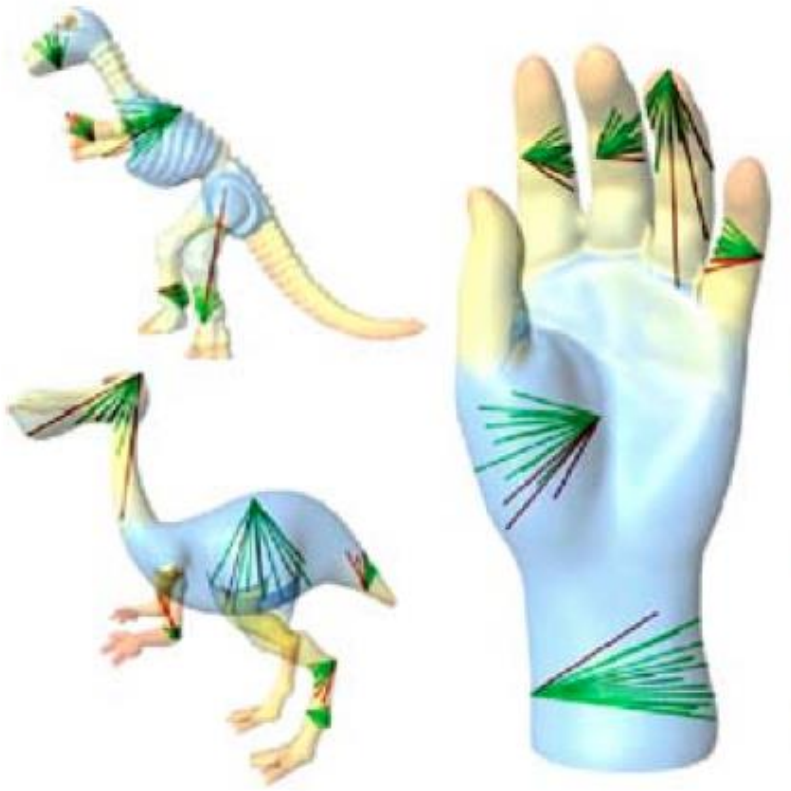
$$f : \Gamma \rightarrow \mathbb{F}$$

Make samples better distinguishable.



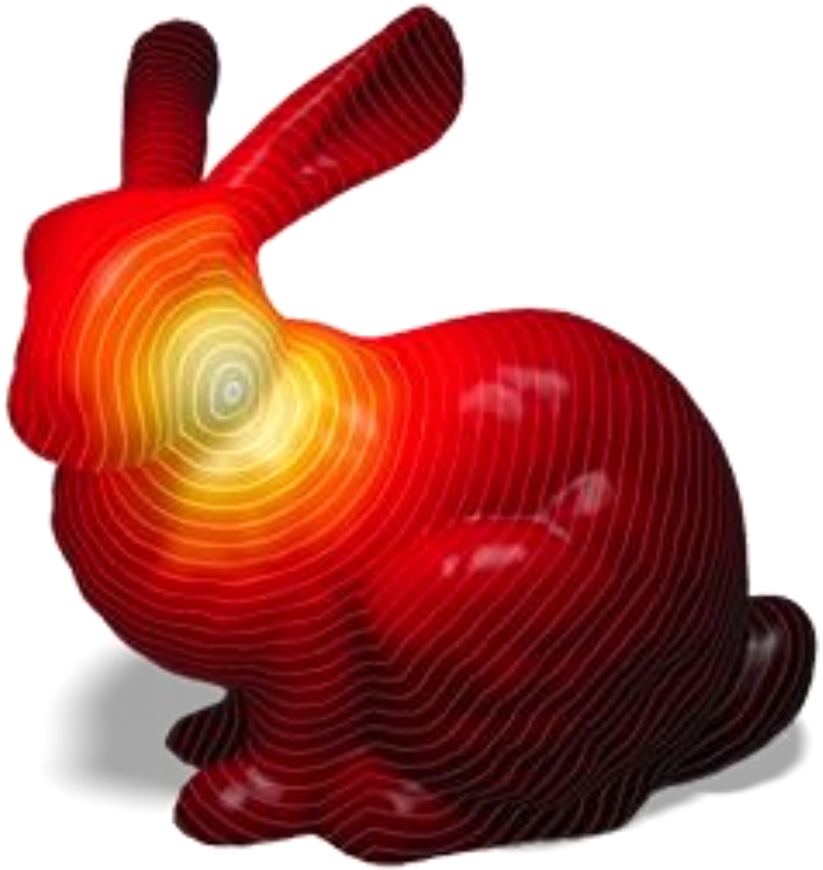
Feature Extraction: Local Diameter

$$f : \Gamma \rightarrow \mathbb{F}$$

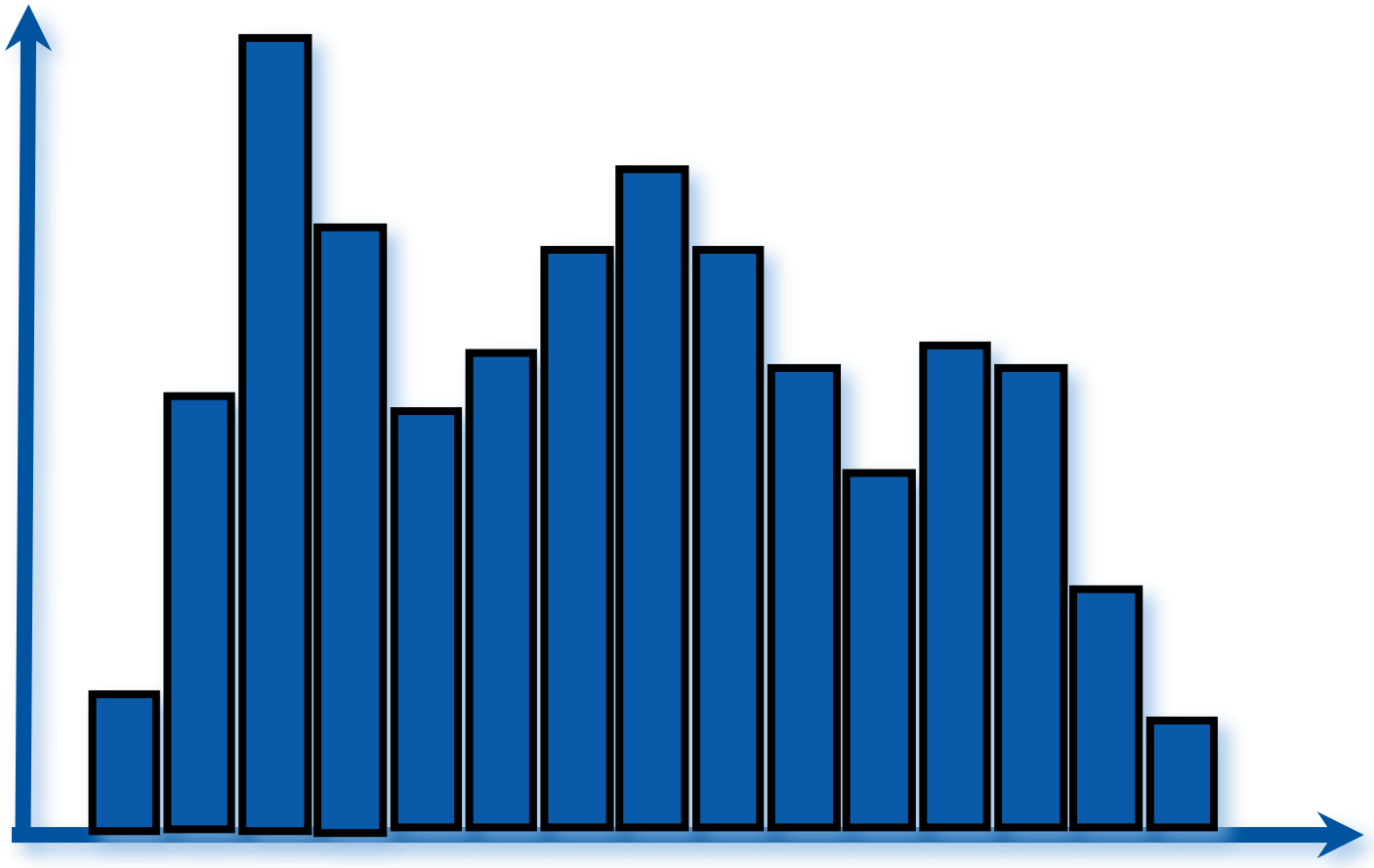


Feature Extraction: Geodesic Distance

$$f : \Gamma \rightarrow \mathbb{F}$$



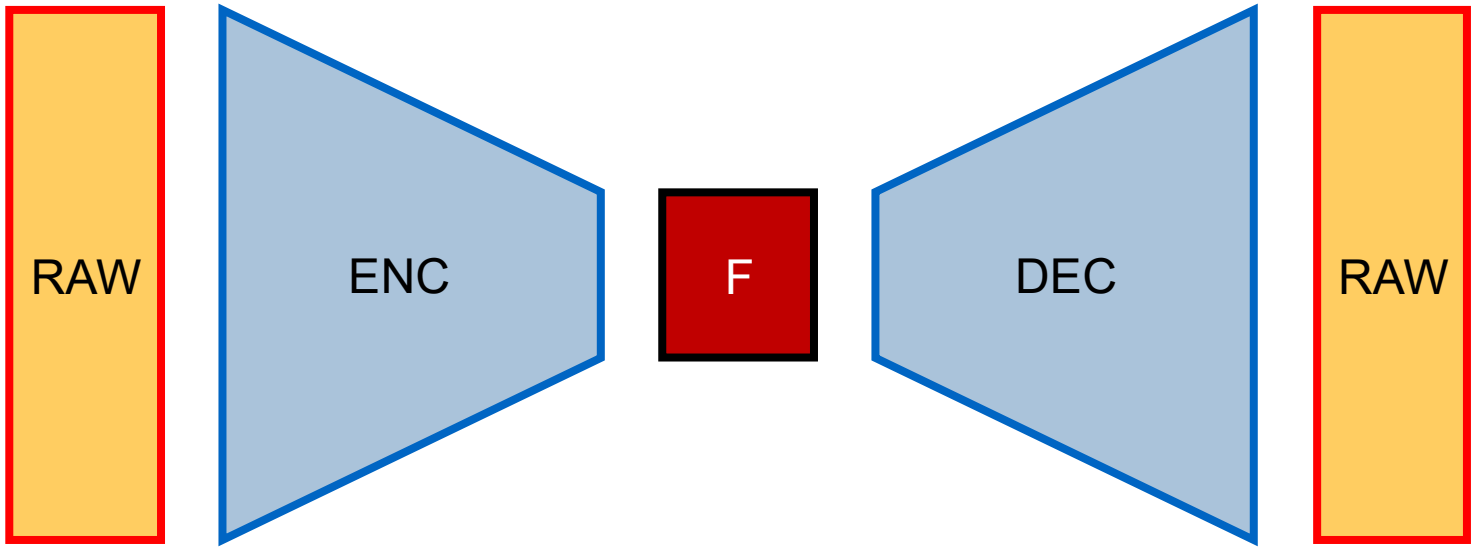
Feature Aggregation: Feature Histogram



Representation Learning (Feature Learning)

Train a neural network („end-to-end“) instead of *handcrafting* features

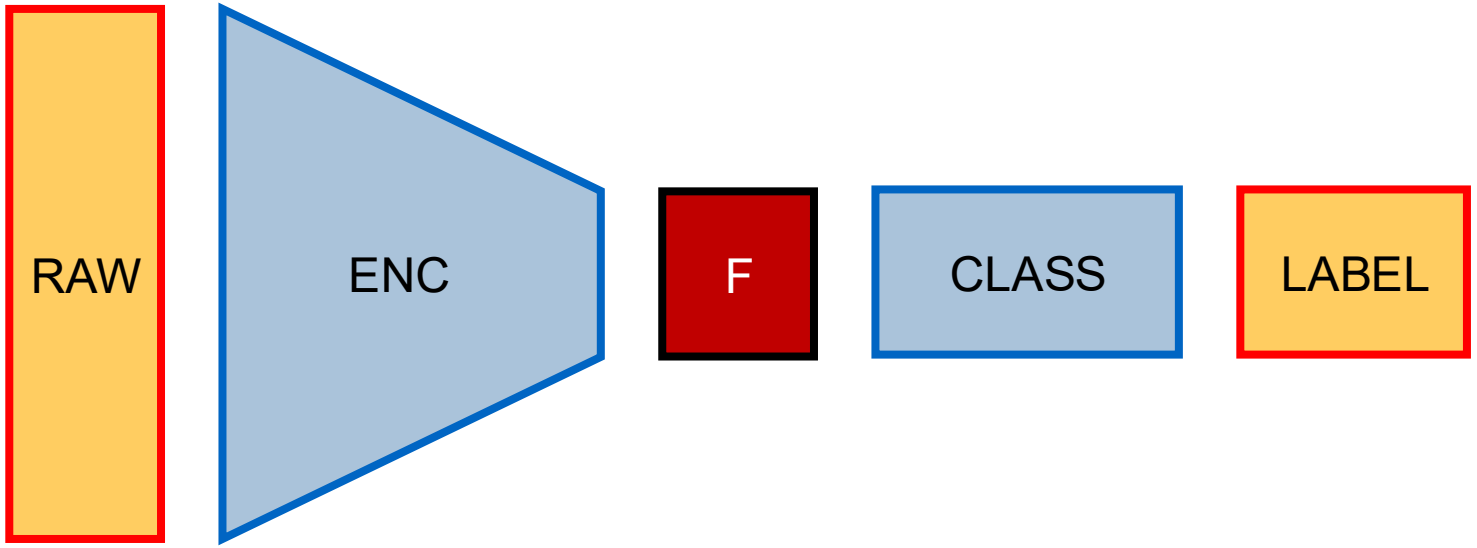
→ unsupervised setting (general purpose, e.g. Autoencoder)



Representation Learning (Feature Learning)

Train a neural network („end-to-end“) instead of *handcrafting* features

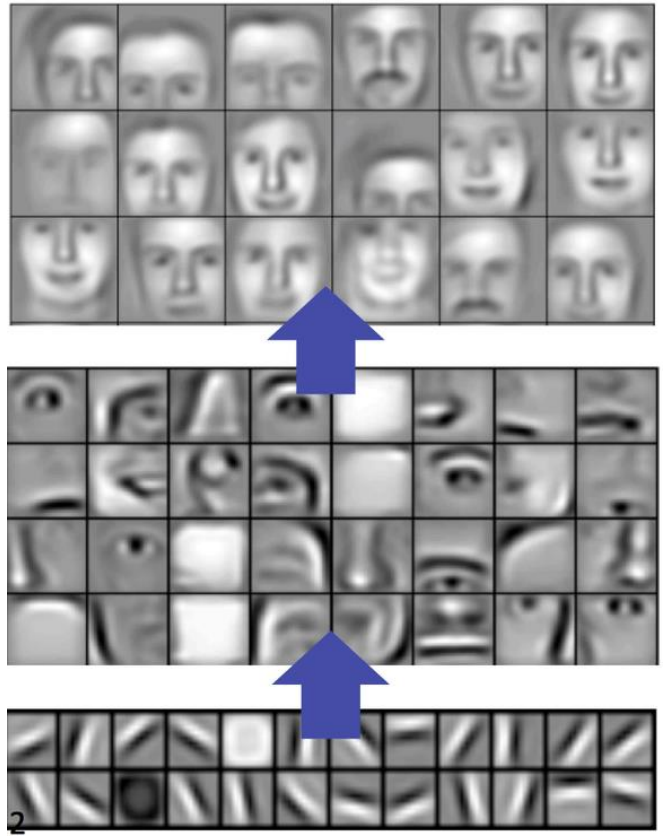
→ supervised setting (features for a special purpose)



Representation Learning (Feature Learning)

Train a neural network („end-to-end“) instead of *handcrafting* features

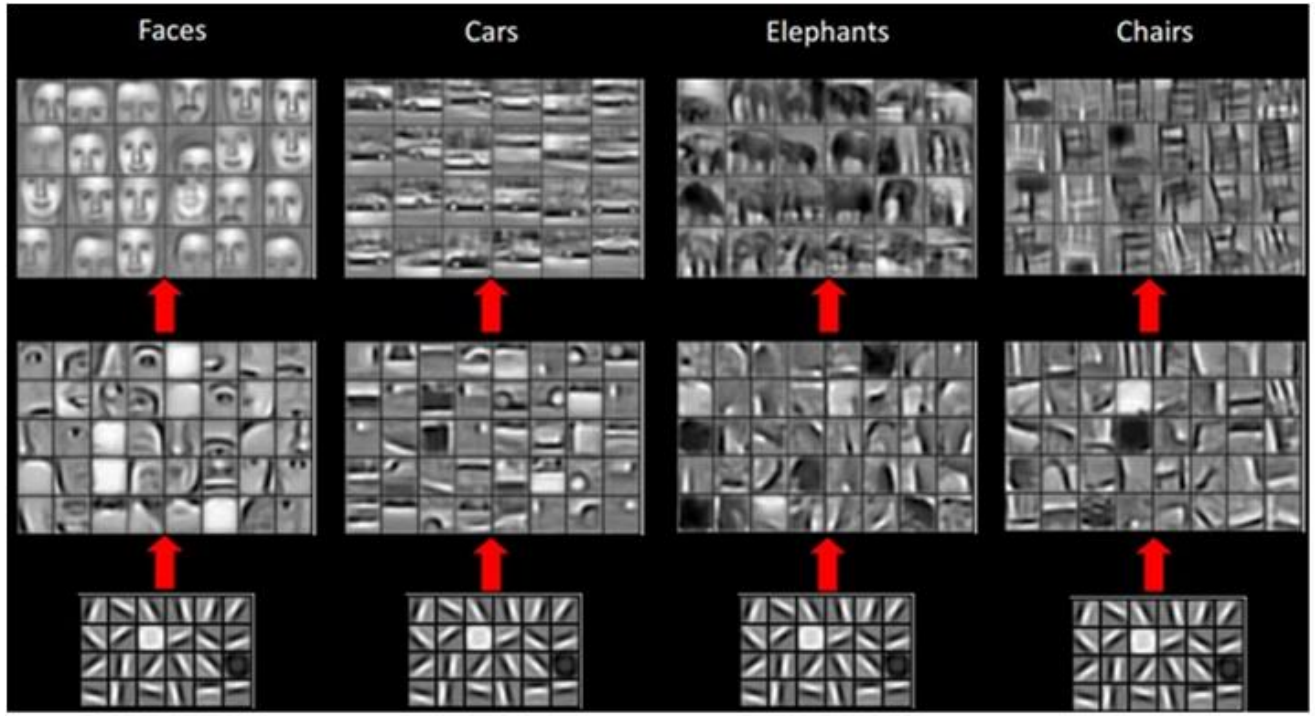
→ supervised setting (features for a special purpose)



Representation Learning (Feature Learning)

Train a neural network („end-to-end“) instead of *handcrafting* features

→ supervised setting (features for a special purpose)



Data Analysis: Outline

- Types of Data
- **Comparing Data**
  - feature spaces
  - **similarity and dissimilarity measures**
  - comparing histograms and distributions
- Dimensionality Reduction
- Clustering

Dissimilarity measures: Dissimilarity Function

$\mathbb{X}$: some set

dissimilarity function: $\delta: \mathbb{X} \times \mathbb{X} \rightarrow \mathbb{R}_{\geq 0}$

- Reflexivity: $\forall x \in \mathbb{X}: \delta(x, x) = 0$
- Non-negativity: $\forall x, y \in \mathbb{X}: \delta(x, y) \geq 0$
- Symmetry: $\forall x, y \in \mathbb{X}: \delta(x, y) = \delta(y, x)$

Dissimilarity measures: Metric Distance Function

$\mathbb{X}$: some set

metric distance function: $\delta: \mathbb{X} \times \mathbb{X} \rightarrow \mathbb{R}_{\geq 0}$

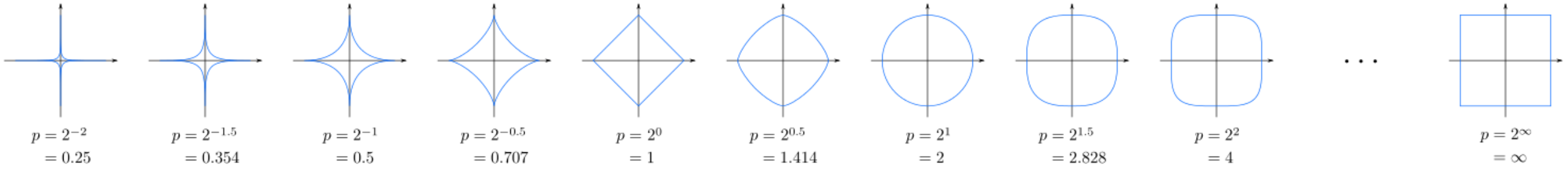
- Reflexivity: $\forall x \in \mathbb{X}: \delta(x, x) = 0$
- Non-negativity: $\forall x, y \in \mathbb{X}: \delta(x, y) \geq 0$
- Symmetry: $\forall x, y \in \mathbb{X}: \delta(x, y) = \delta(y, x)$
- Triangle Inequality: $\forall x, y, z \in \mathbb{X}: \delta(x, y) \leq \delta(x, z) + \delta(z, y)$

Minkowski Distance

$$L_p(X,Y) = \left(\sum_i |X_i - Y_i|^p \right)^{\frac{1}{p}}$$

Metric for $p \geq 1$

- $p = 1$: Manhattan distance
- $p = 2$: Euclidean distance
- $p = \infty$: Chebyshev/Maximum distance



[Wikipedia]

Dissimilarity Measures

(Metric) distance functions allow:

- domain experts to model their notion of dissimilarity (can also be learned by a neural network)
- database experts to design efficient query processing approaches (especially if the triangle inequality can be used)

Similarity Functions

$\mathbb{X}$: some set

similarity function: $s: \mathbb{X} \times \mathbb{X} \rightarrow \mathbb{R}$

- Symmetry: $\forall x, y \in \mathbb{X}: s(x, y) = s(y, x)$
- Maximum Self-Similarity: $\forall x, y \in \mathbb{X}: s(x, x) \geq s(x, y)$

From distance functions to similarity functions

We can transform any distance function δ to a similarity function s with any monotonically decreasing function $g: \mathbb{R} \to \mathbb{R}$

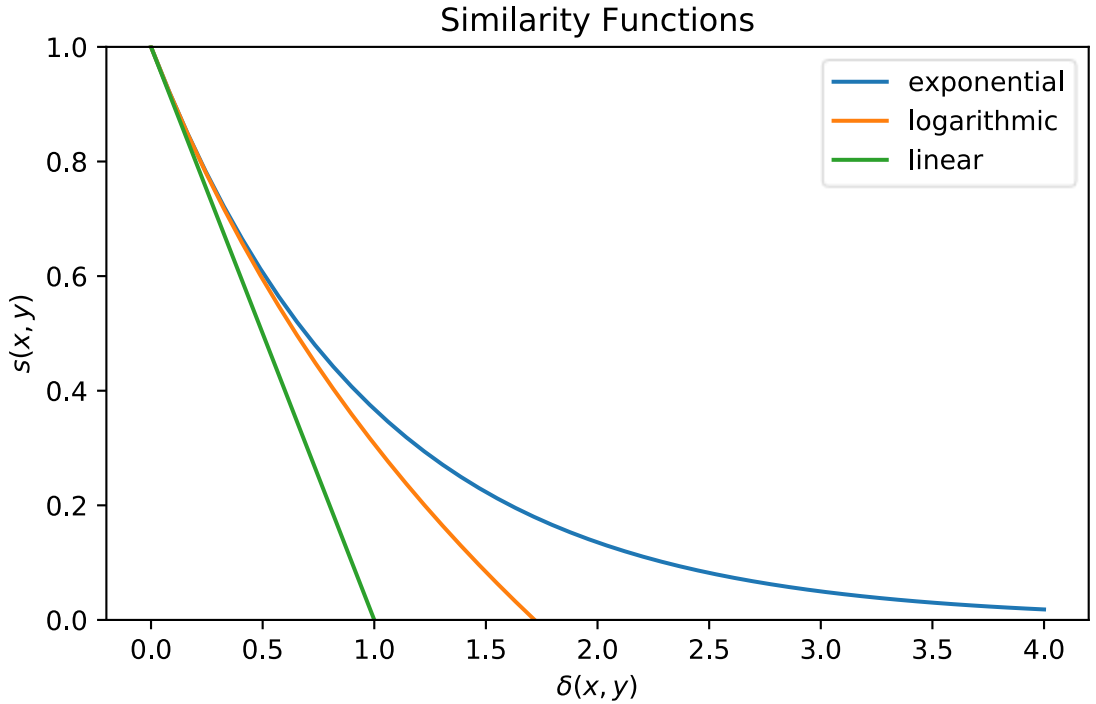
$$\forall x, y \in \mathbb{X}: s(x, y) = g\big(\delta(x, y)\big)$$

Similarity Function Examples

Linear: $s(x,y) = 1 - \delta(x,y)$

Logarithmic: $s(x,y) = 1 - \log\big(1 + \delta(x,y)\big)$

Exponential: $s(x,y) = e^{-\delta(x,y)}$



Similarity Measures: Dot Product Measure

Used in many other similarity measures

$$x, y \in \mathbb{R}^d : s_d(x, y) = x^T y = \sum_{i=1}^{d} x_i y_i = \|x\| \cdot \|y\| \cos\theta$$

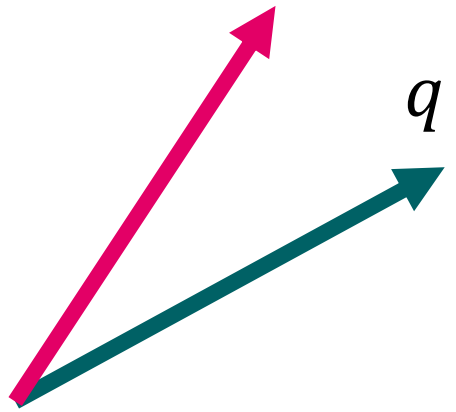
- Direction of vector corresponds to properties of the object
- Length of vector corresponds to intensity of properties



Dot Product Measure: Angle monotonicity

Given vectors of the same length:

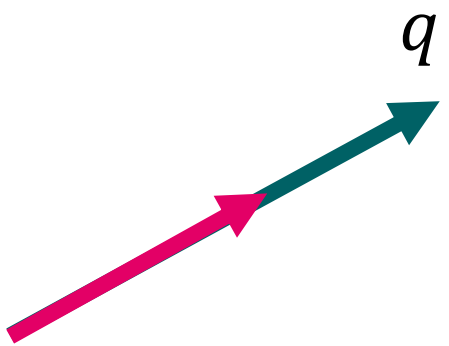
- the *smaller* the angle between the vector and q, the *higher* the similarity



Dot Product Measure: Radial monotonicity

Given vectors with the same direction:

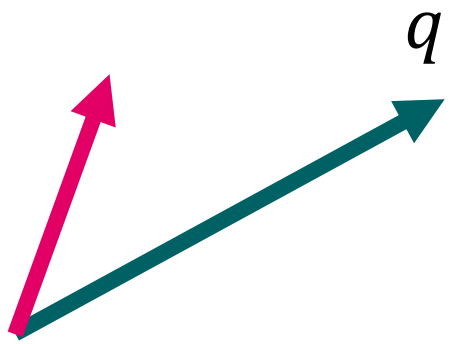
- the *longer* the vector q, the *higher* the similarity



Dot Product Measure: component-wise monotonicity

Changing the value of components dimension-wise will change the similarity accordingly.

It is possible to achieve arbitrarily high similarity by increasing the intensity of a single property

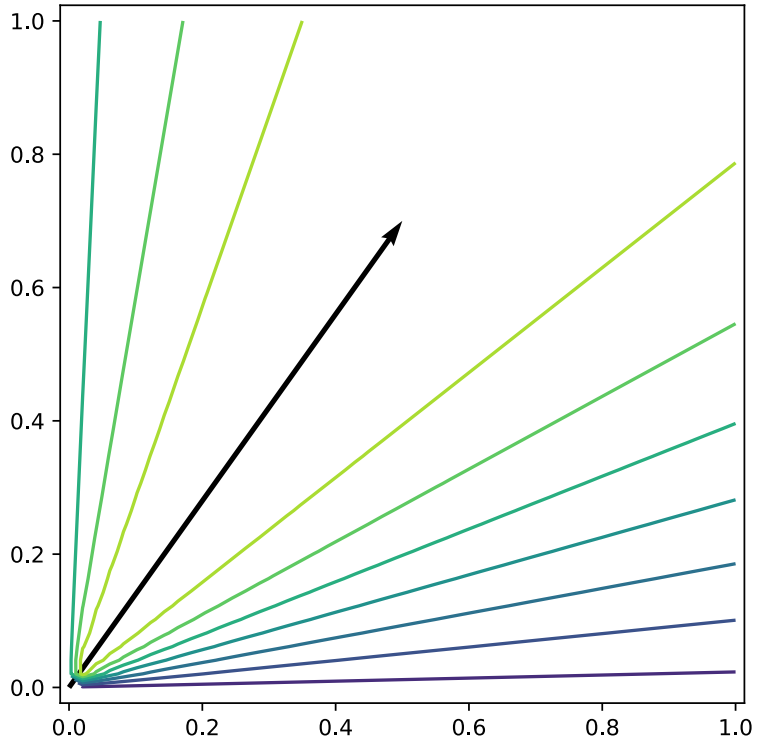


Similarity Measures: Cosine Measure

Normalized dot product

$$x, y \in \mathbb{R}^d : s_c(x, y) = \frac{x^T y}{\|x\| \cdot \|y\|}$$

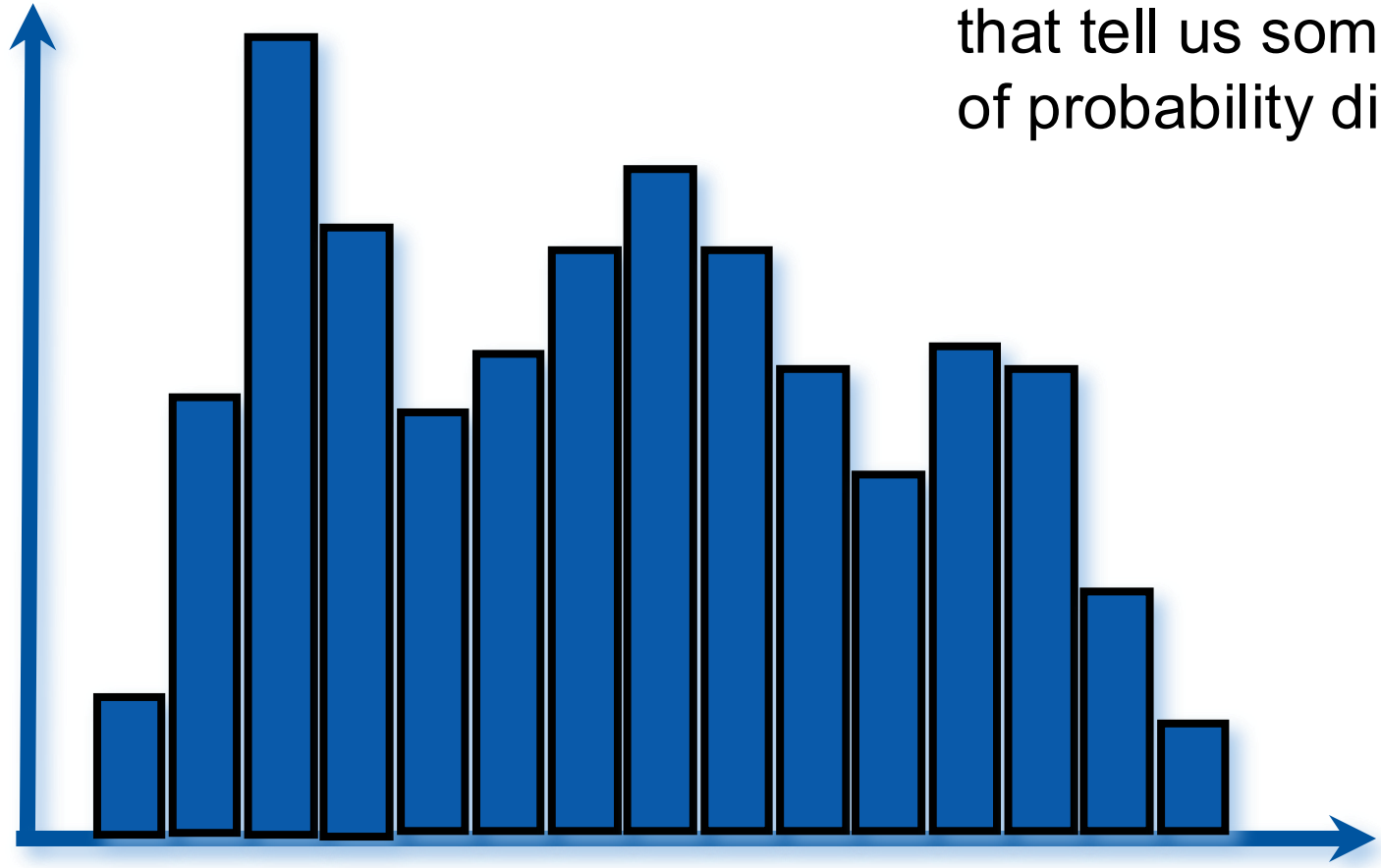
Only the angle defines the similarity



Data Analysis: Outline

- Types of Data
- **Comparing Data**
  - feature spaces
  - similarity and dissimilarity measures
  - **comparing histograms and distributions**
- Dimensionality Reduction
- Clustering

Feature Aggregation: Feature Histogram

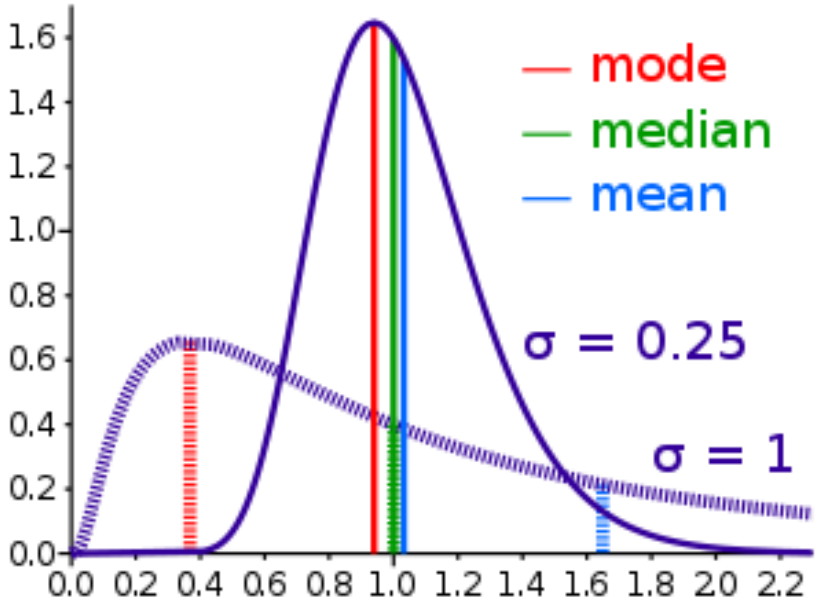


What characteristics can we compute that tell us something about the shape of probability distributions/histograms?

Central Tendency

Given data/histogram $\mathrm{X} = (x_1, \ldots, x_n)$

(Arithmetic) **Mean** $\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$



[Wikipedia]

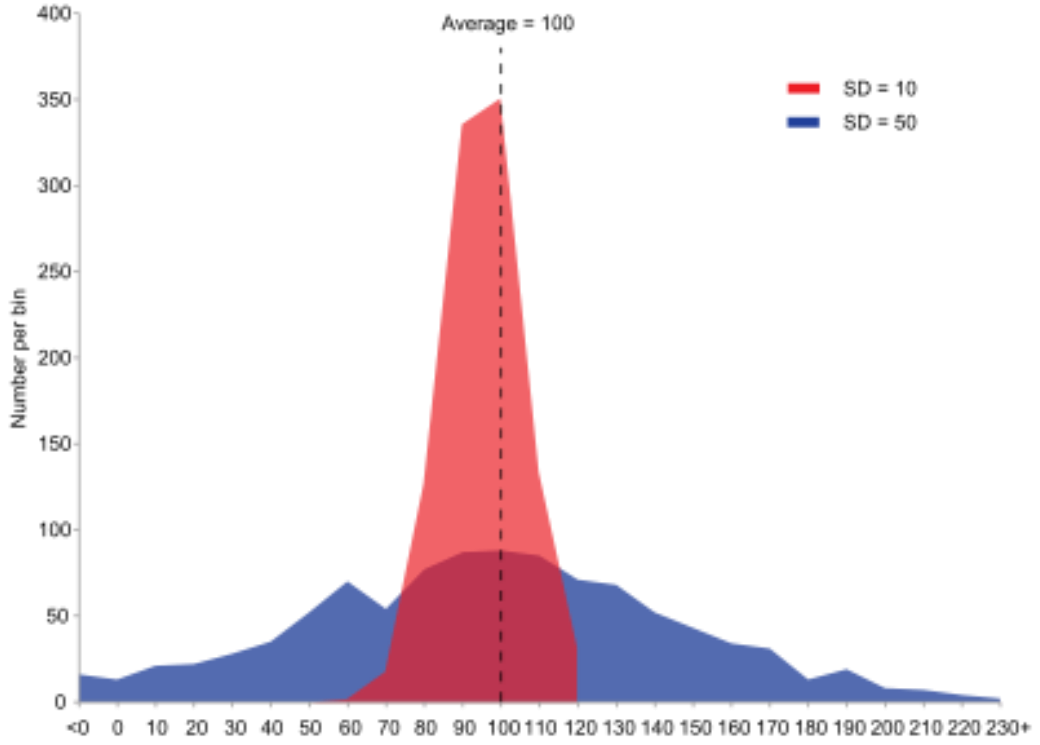
Median: "Middle of the dataset": $\frac{x_{\lfloor \frac{n+1}{2} \rfloor} + x_{\lceil \frac{n+1}{2} \rceil}}{2}$, for sorted x_i

Mode: most frequent value in dataset (highest value)

Spread

Variance: $Var(X) = \sigma^2 = E[(X - \mu)^2] = \frac{\sum_{i=1}^{n}(x_i - \mu)^2}{n}$

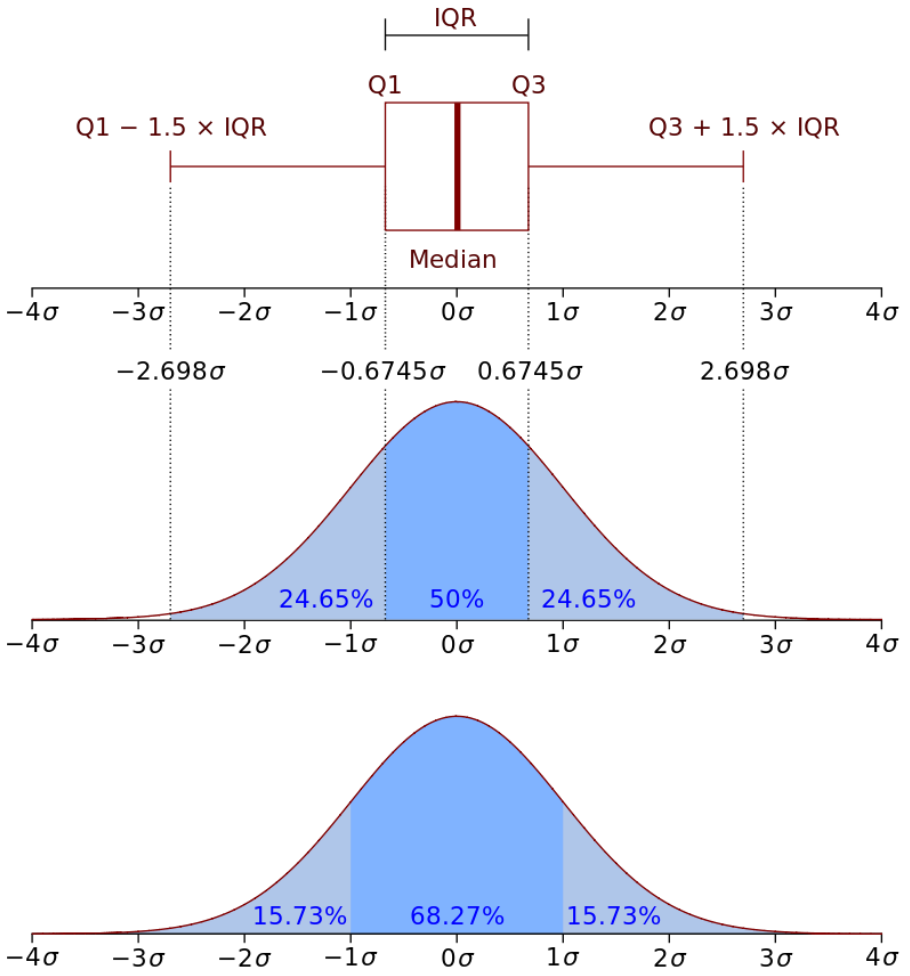
Standard deviation: $\sigma = \sqrt{Var(X)}$



[Wikipedia]

Spread: Quartiles

Q1: first quartile: 25th percentile
Q2: median: 50th percentile
Q3: third quartile: 75th percentile

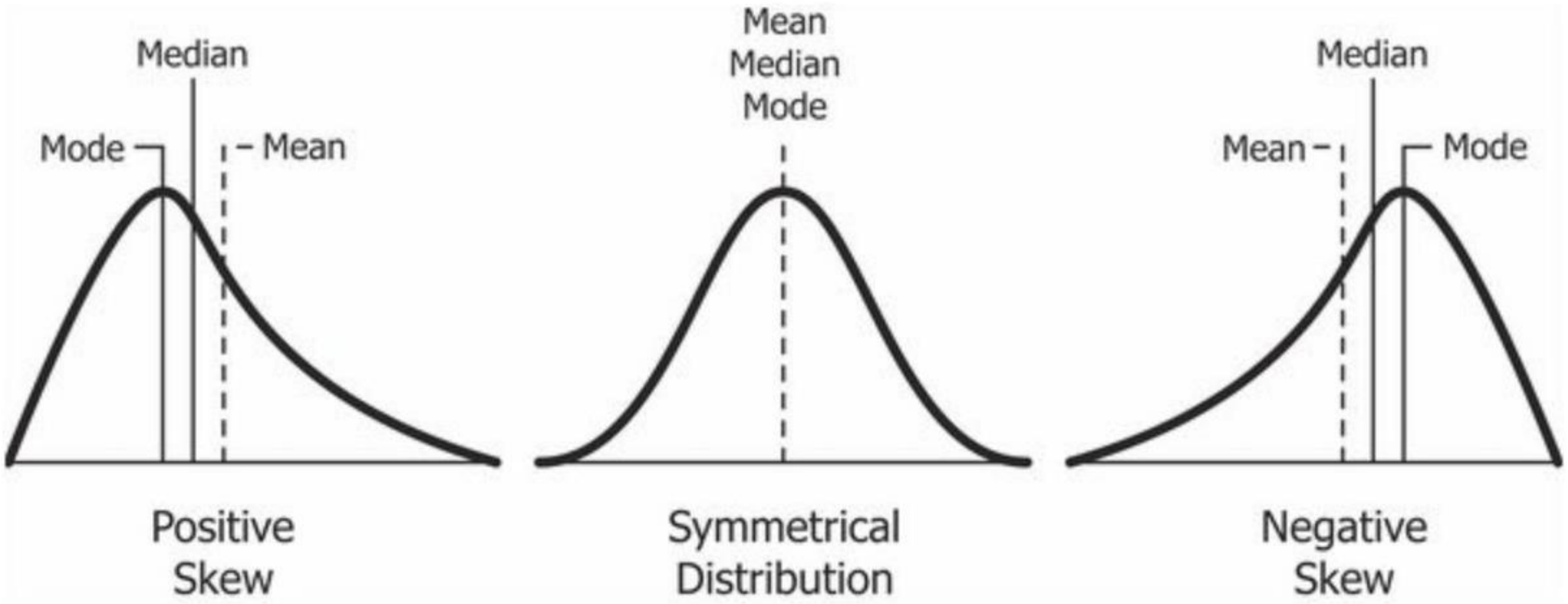


„68-95-99.7“ rule

[Wikipedia]

Skewness

Third standardized moment $\tilde{\mu}_3 = E\left[\left(\frac{X-\mu}{\sigma}\right)^3\right] = \frac{\mu_3}{\sigma^3} = \frac{E[(X-\mu)^3]}{\sigma^3}$

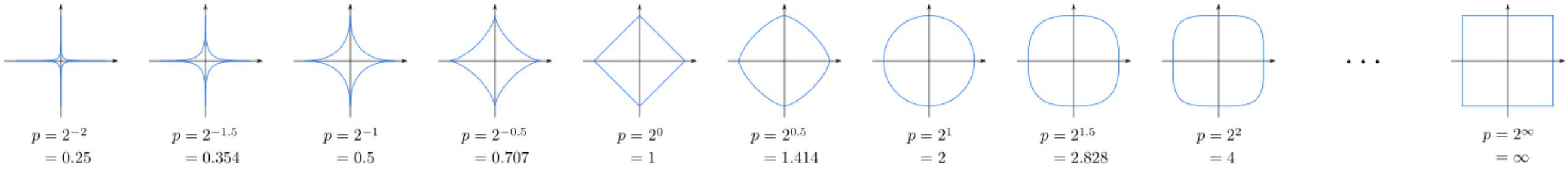


Minkowski Distance (reprise)

$$L_p(X,Y) = \left(\sum_{f \in R} |X(f) - Y(f)|^p \right)^{\frac{1}{p}}$$

Metric for $p \geq 1$

- $p = 1$: Manhattan distance
- $p = 2$: Euclidean distance
- $p = \infty$: Chebyshev/Maximum distance



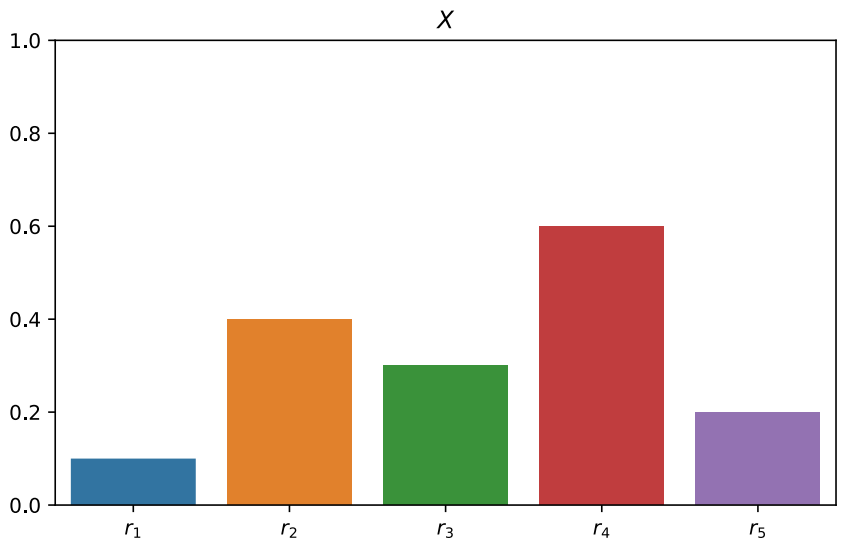
[Wikipedia]

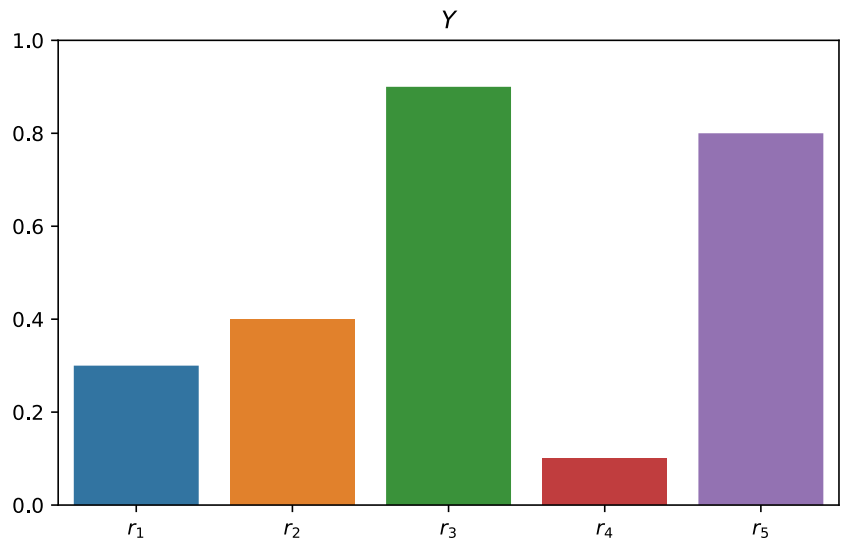
Visual Computing Institute | Dr. Isaak Lim
Computer Graphics and Multimedia
Data Analysis and Visualization

Visual Computing Institute | RWTH AACHEN UNIVERSITY

Distance Functions for Feature Histograms

Given two feature histograms $X, Y \in \mathbb{H}_R$ with bins $R = \{r_1, r_2, r_3, r_4, r_5\}$



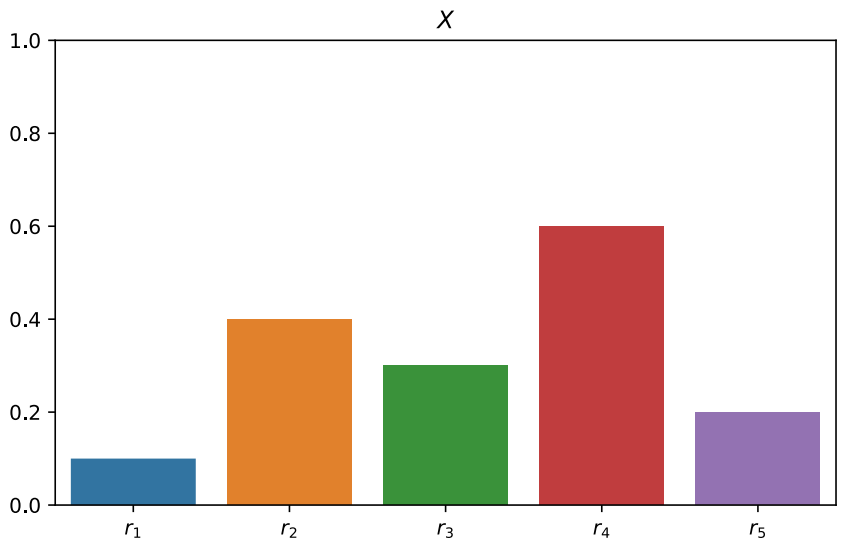


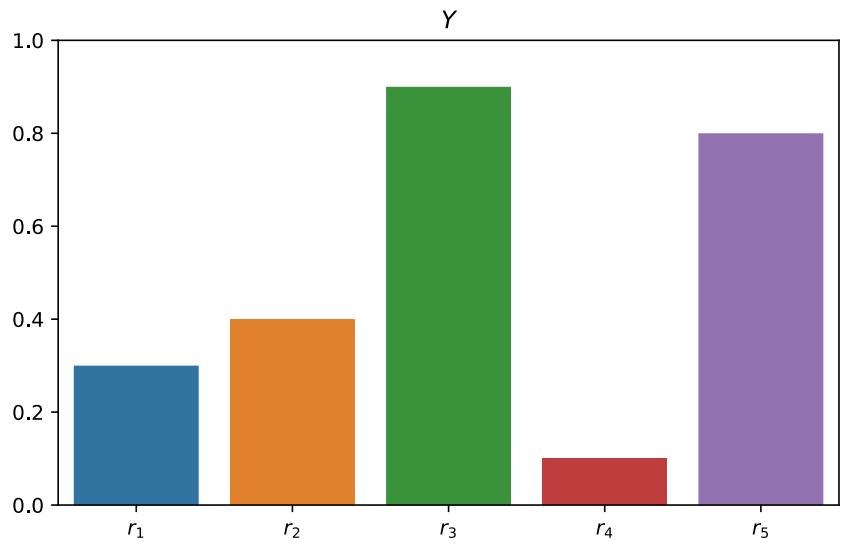
Hamming Distance

Count the number of bins with different weights

$$HammingDist(X,Y) = \sum_{f \in R} \begin{cases} 1 & if\ X(f) \neq Y(f) \\ 0 & else \end{cases}$$

Hamming distance is a metric on feature histograms

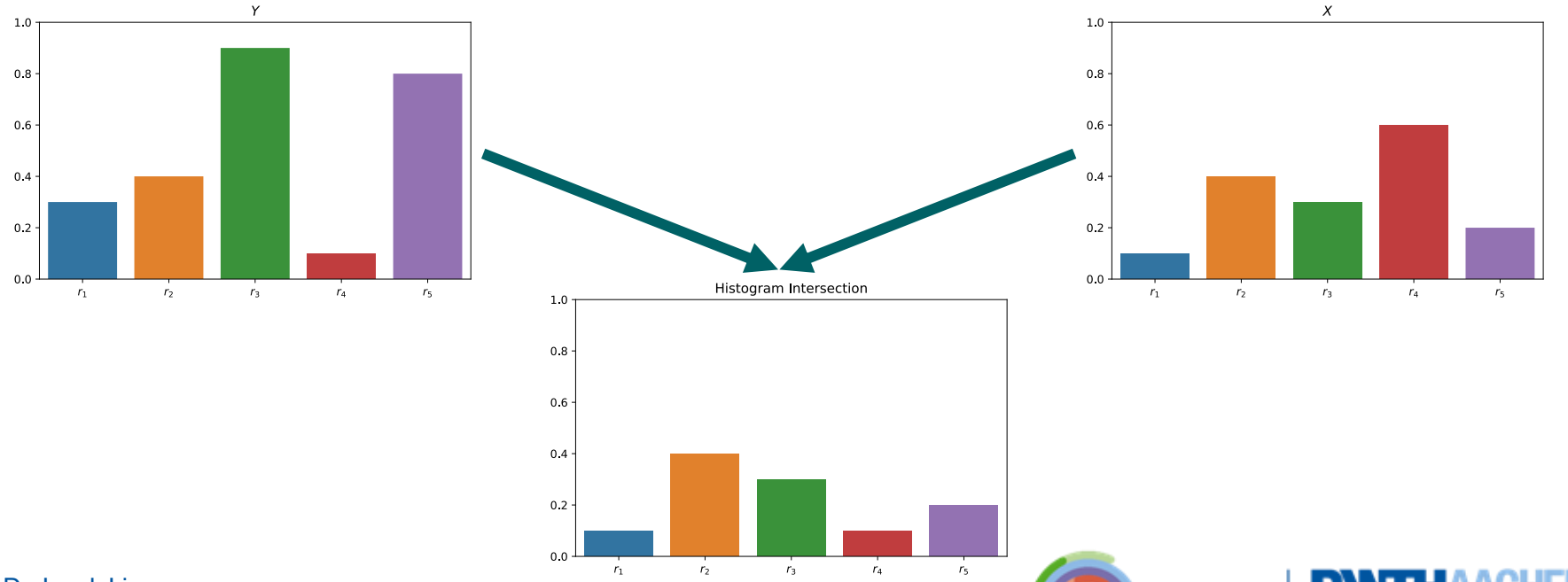




Histogram Intersection

Minimal value of weights that two histograms share

$$HI(X,Y) = \frac{\sum_{f \in R} \min\{X(f), Y(f)\}}{\sum_{f \in R} Y(f)}$$

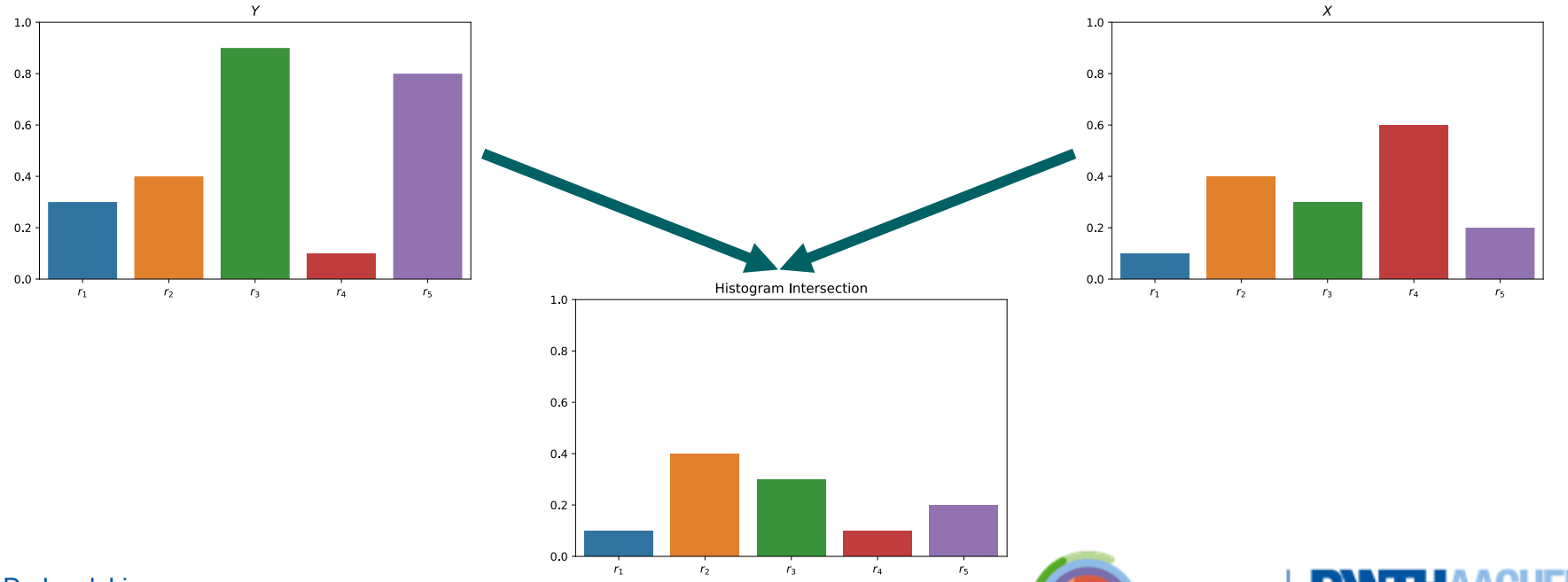


Not symmetric

Histogram Intersection over Union (IoU)

$$\boldsymbol{IoU(X,Y)} = \frac{\sum_{f\in R}\min\{X(f),Y(f)\}}{\sum_{f\in R}Y(f)+\sum_{f\in R}X(f)-\sum_{f\in R}\min\{X(f),Y(f)\}} = \frac{\sum_{f\in R}\boldsymbol{min}\{\boldsymbol{X(f)},\boldsymbol{Y(f)}\}}{\sum_{f\in R}\boldsymbol{max}\{\boldsymbol{X(f)},\boldsymbol{Y(f)}\}}$$

$0 \leq IoU(X,Y) \leq 1$, (if both histograms are empty defined as 1)



Symmetric

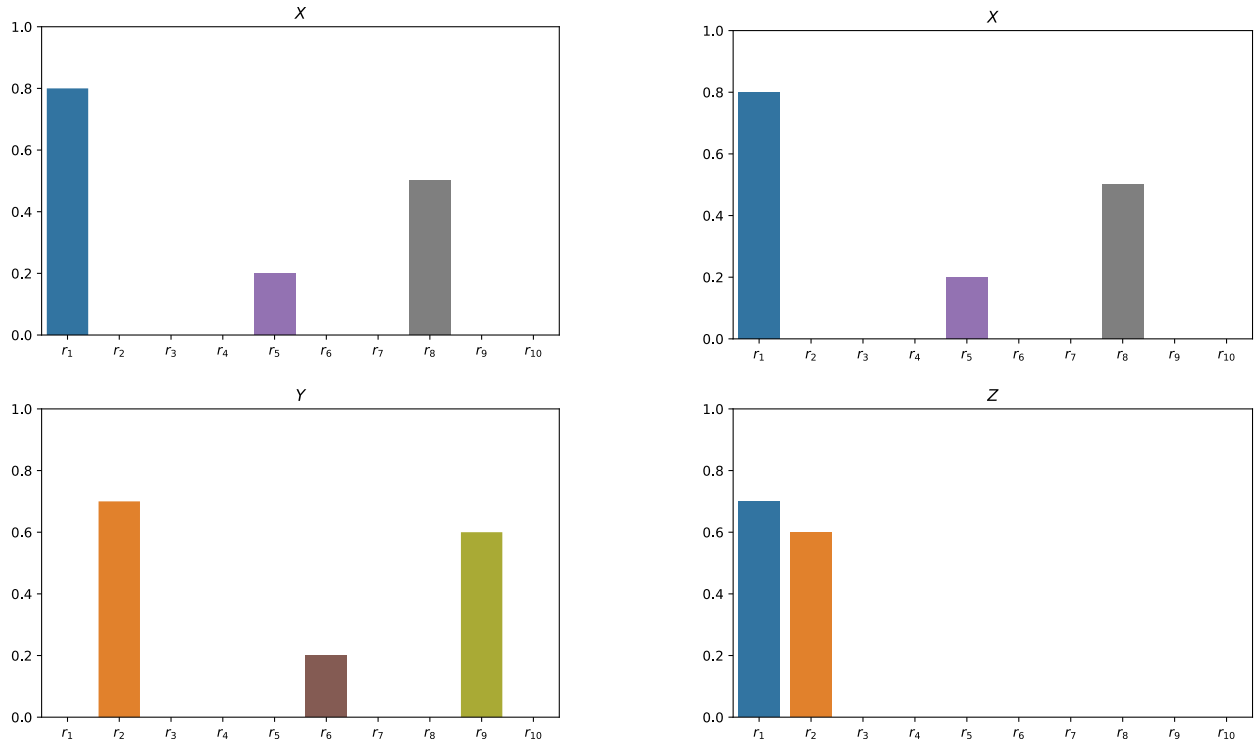
Visual Computing Institute | Dr. Isaak Lim
Computer Graphics and Multimedia
Data Analysis and Visualization

Visual Computing Institute
RWTH AACHEN UNIVERSITY

Issues with Bin-by-Bin Distance Functions

Define distance by only looking at single dimensions (bins)
Neighboring dimensions (bins) are ignored

Example: $L_p(X,Y) \geq L_p(X,Z)$

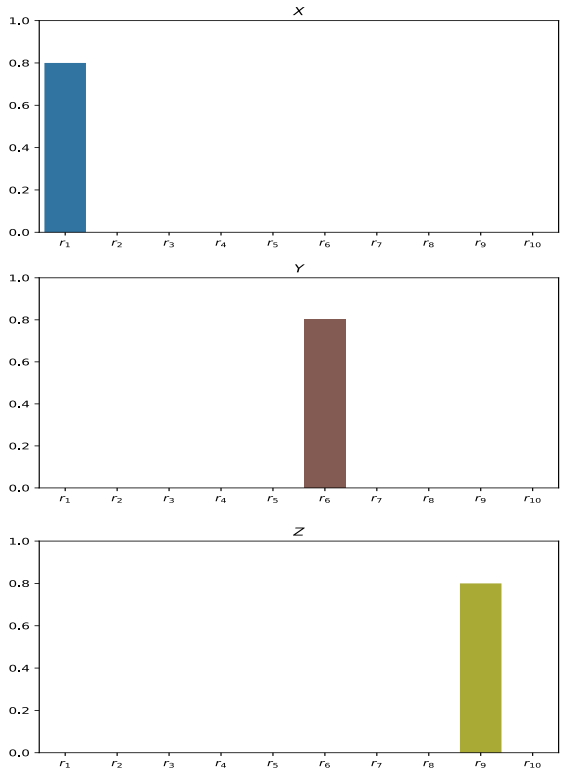


Visual Computing Institute
RWTH AACHEN UNIVERSITY

Issues with Bin-by-Bin Distance Functions

Define distance by only looking at single dimensions (bins)
Neighboring dimensions (bins) are ignored

Example: $L_p(X,Y) = L_p(X,Z) = L_p(Y,Z)$



Visual Computing Institute | RWTH AACHEN UNIVERSITY

Cross-Bin Distance Functions

Introduce weighting of pairs of bins (dimensions)

Weighting/influence can be modeled with similarity function

Quadratic Form Distance

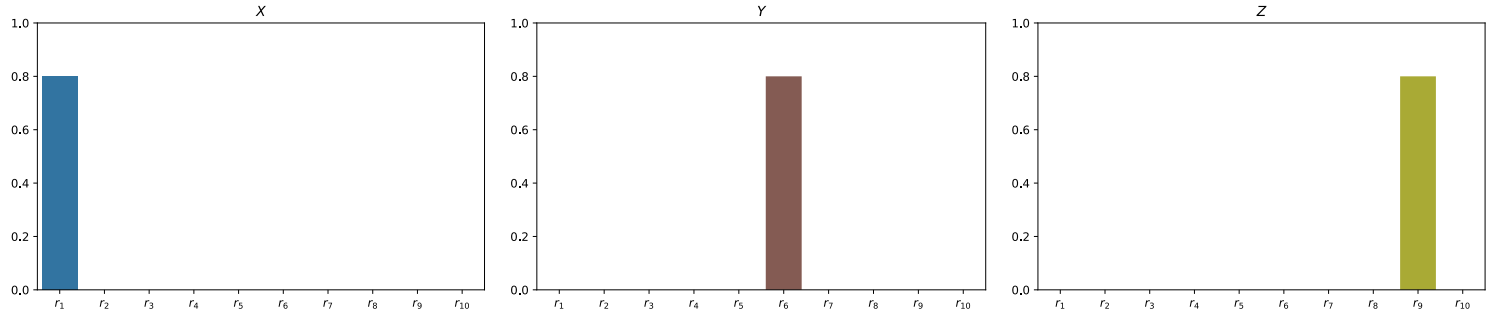
$$QFD_s(X,Y) = \sqrt{\sum_{f\in R}\sum_{g\in R}\big(X(f)-Y(f)\big)\cdot s(f,g)\cdot\big(X(g)-Y(g)\big)}$$

Define similarity matrix $S = \begin{pmatrix} s(r_1,r_1) & \cdots & s(r_1,r_R) \\ \vdots & \ddots & \vdots \\ s(r_R,r_1) & \cdots & s(r_R,r_R) \end{pmatrix}$

$$QFD_s(X,Y) = \sqrt{(x-y)\cdot S\cdot(x-y)^T}$$

- $x = \big(X(r_1),\ldots,X(r_R)\big)$
- $y = \big(Y(r_1),\ldots,Y(r_R)\big)$

Quadratic Form Distance: Example

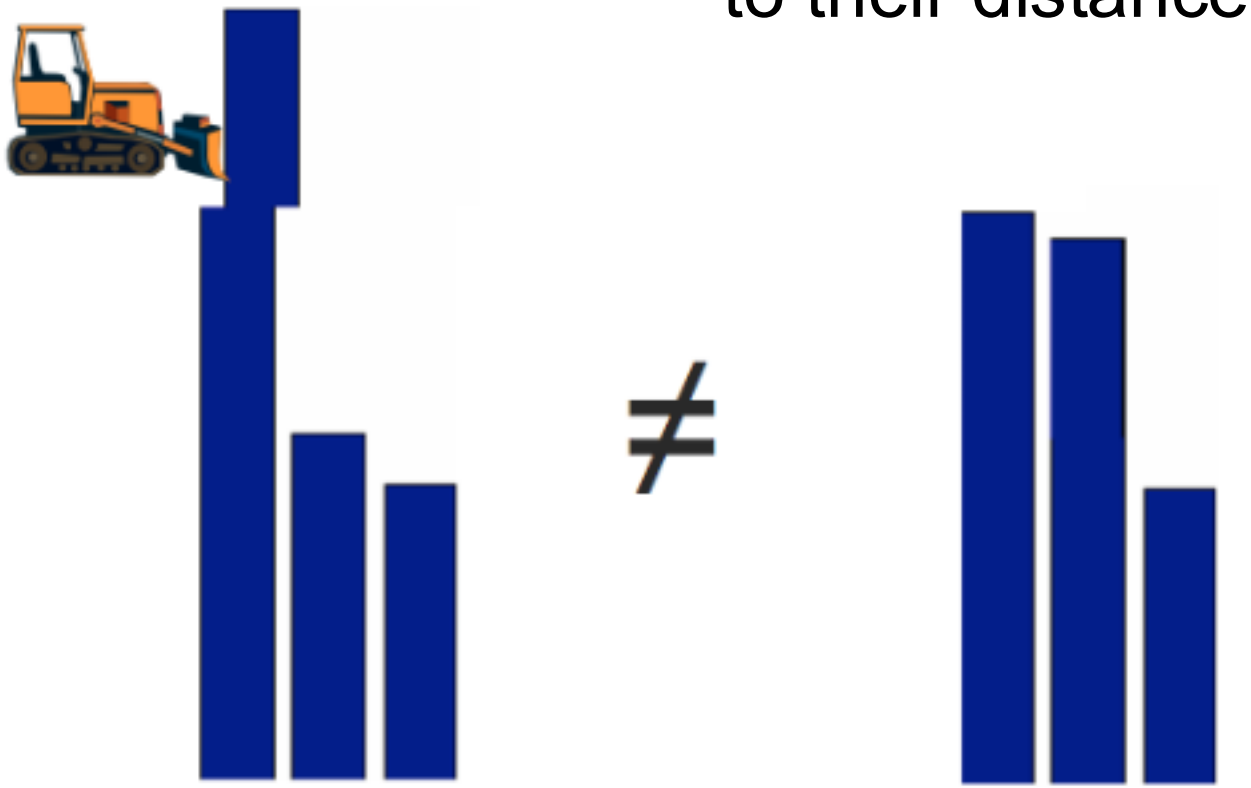


$$S = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0.2 & 0 & 0 & 0.2 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0.2 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0.6 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0.2 & 0 & 0 & 0 & 0 & 0.6 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

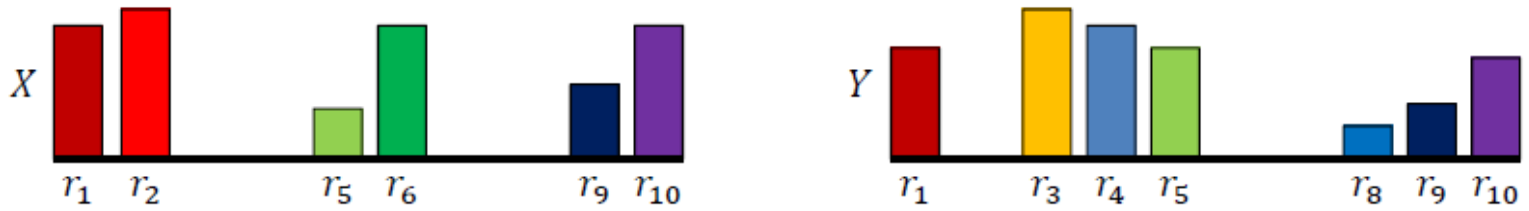
$$QFD_s(X, Y) \approx 1.01$$
$$QFD_s(X, Z) \approx 1.01$$
$$QFD_s(Y, Z) \approx 0.72$$

Earth Mover's Distance / First Order Wasserstein Distance

Distance between distributions … bin similarity proportional to their distance



Earth Mover's Distance … Optimal Transport



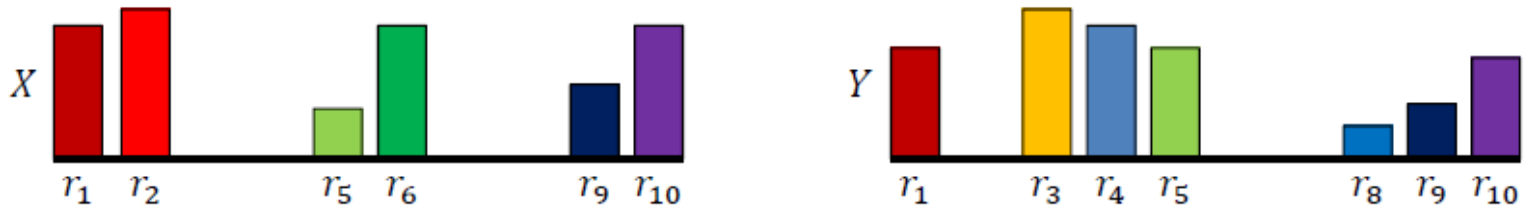
Non-empty bins:

- $R_X = \{r_1, r_2, r_5, r_6, r_9, r_{10}\}$
- $R_Y = \{r_1, r_3, r_4, r_5, r_8, r_9, r_{10}\}$

Earth piles (sources) R_X with capacities $X(r_i), r_i \in R_X$

Earth holes (sinks) R_Y with capacities $Y(r_i), r_i \in R_Y$

Earth Mover's Distance … Optimal Transport



Amount of earth that is moved: **flow** $f: R_X \times R_Y \rightarrow \mathbb{R}$

Cost of moving earth: **ground distance** $\delta: R_X \times R_Y \rightarrow \mathbb{R}$

$$EMD(X,Y) = \min_{\{f \mid f: R_X \times R_Y \rightarrow \mathbb{R}\}} \frac{\sum_{g \in R_X} \sum_{h \in R_Y} f(g,h) \cdot \delta(g,h)}{\min\{\sum_{g \in R_X} X(g), \sum_{h \in R_Y} Y(h)\}}$$

Visual Computing Institute | RWTH AACHEN UNIVERSITY

Earth Mover's Distance … Optimal Transport

$$EMD(X,Y) = \min_{\{f \mid f: R_X \times R_Y \to \mathbb{R}\}} \frac{\sum_{g \in R_X} \sum_{h \in R_Y} f(g,h) \cdot \delta(g,h)}{\min\{\sum_{g \in R_X} X(g), \sum_{h \in R_Y} Y(h)\}}$$

Subject to:

- Non-negativity: $\forall g \in R_X, h \in R_Y: f(g,h) \geq 0$
- Do not exceed capacity of sources: $\forall g \in R_X: \sum_{h \in R_Y} f(g,h) \leq X(g)$
- Do not exceed capacity of sinks: $\forall h \in R_Y: \sum_{g \in R_X} f(g,h) \leq Y(h)$
- The flow is maximal: $\sum_{g \in R_X} \sum_{h \in R_Y} f(g,h) = \min\{\sum_{g \in R_X} X(g), \sum_{h \in R_Y} Y(h)\}$

Earth Mover's Distance … Optimal Transport

$$EMD(X,Y) = \min_{\{f \mid f: R_X \times R_Y \to \mathbb{R}\}} \frac{\sum_{g \in R_X} \sum_{h \in R_Y} f(g,h) \cdot \delta(g,h)}{\min\{\sum_{g \in R_X} X(g), \sum_{h \in R_Y} Y(h)\}}$$

Can be very expensive to compute

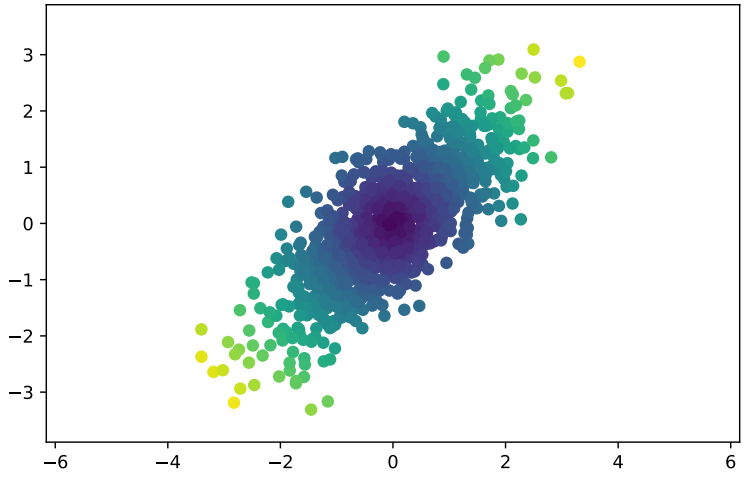
Subject to:

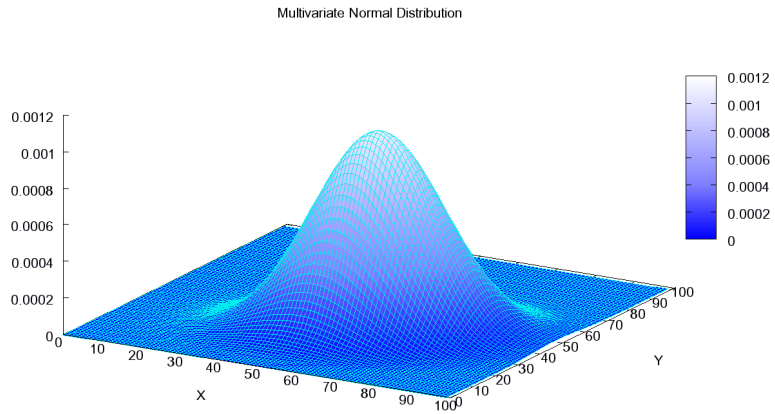
- Non-negativity: $\forall g \in R_X, h \in R_Y: f(g,h) \geq 0$
- Do not exceed capacity of sources: $\forall g \in R_X: \sum_{h \in R_Y} f(g,h) \leq X(g)$
- Do not exceed capacity of sinks: $\forall h \in R_Y: \sum_{g \in R_X} f(g,h) \leq Y(h)$
- The flow is maximal: $\sum_{g \in R_X} \sum_{h \in R_Y} f(g,h) = \min\{\sum_{g \in R_X} X(g), \sum_{h \in R_Y} Y(h)\}$

Beyond Histograms … Probability distributions

View the samples in a datasets in terms of (unknown) distributions that generated them (*that has been fitted to them*)

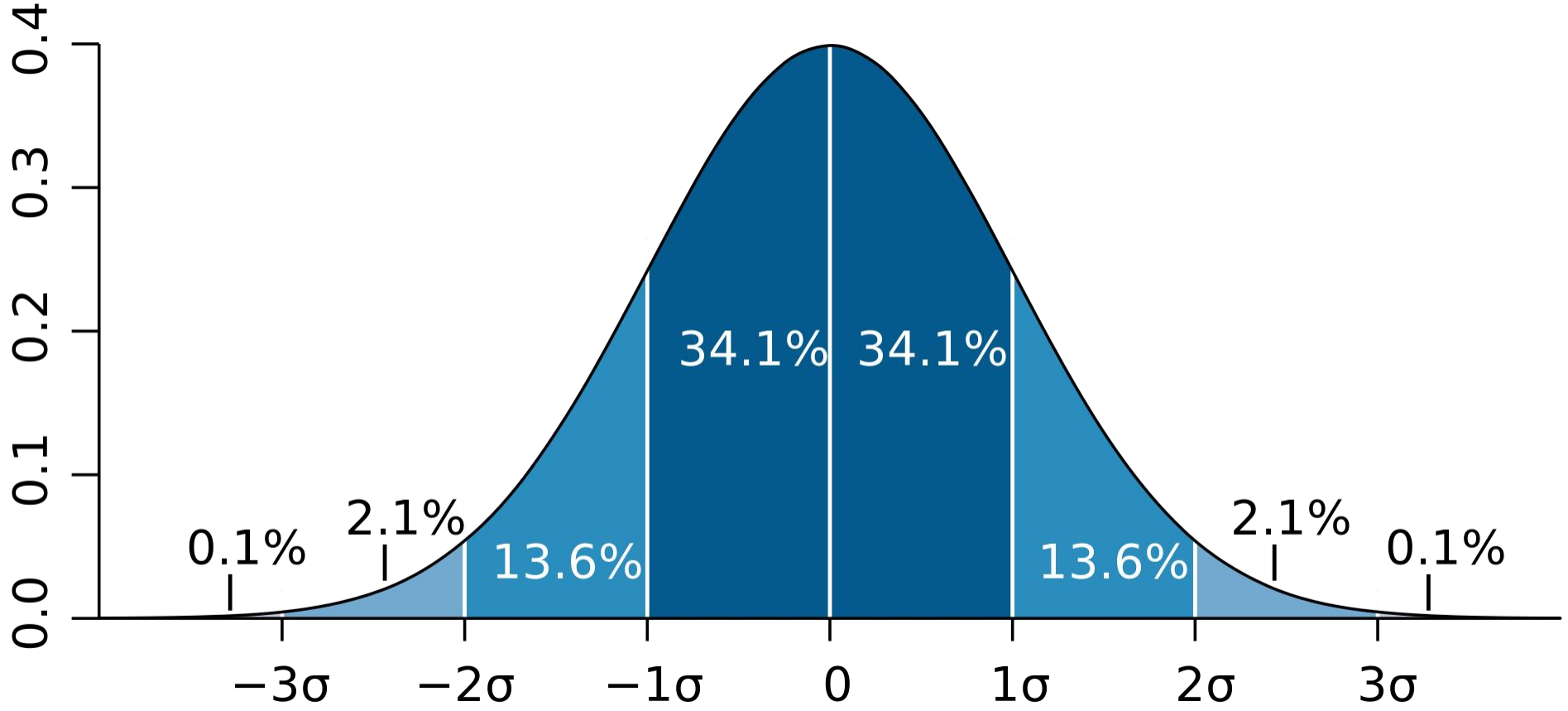
- Distance of new data points to a distribution
- Distance between probability distributions





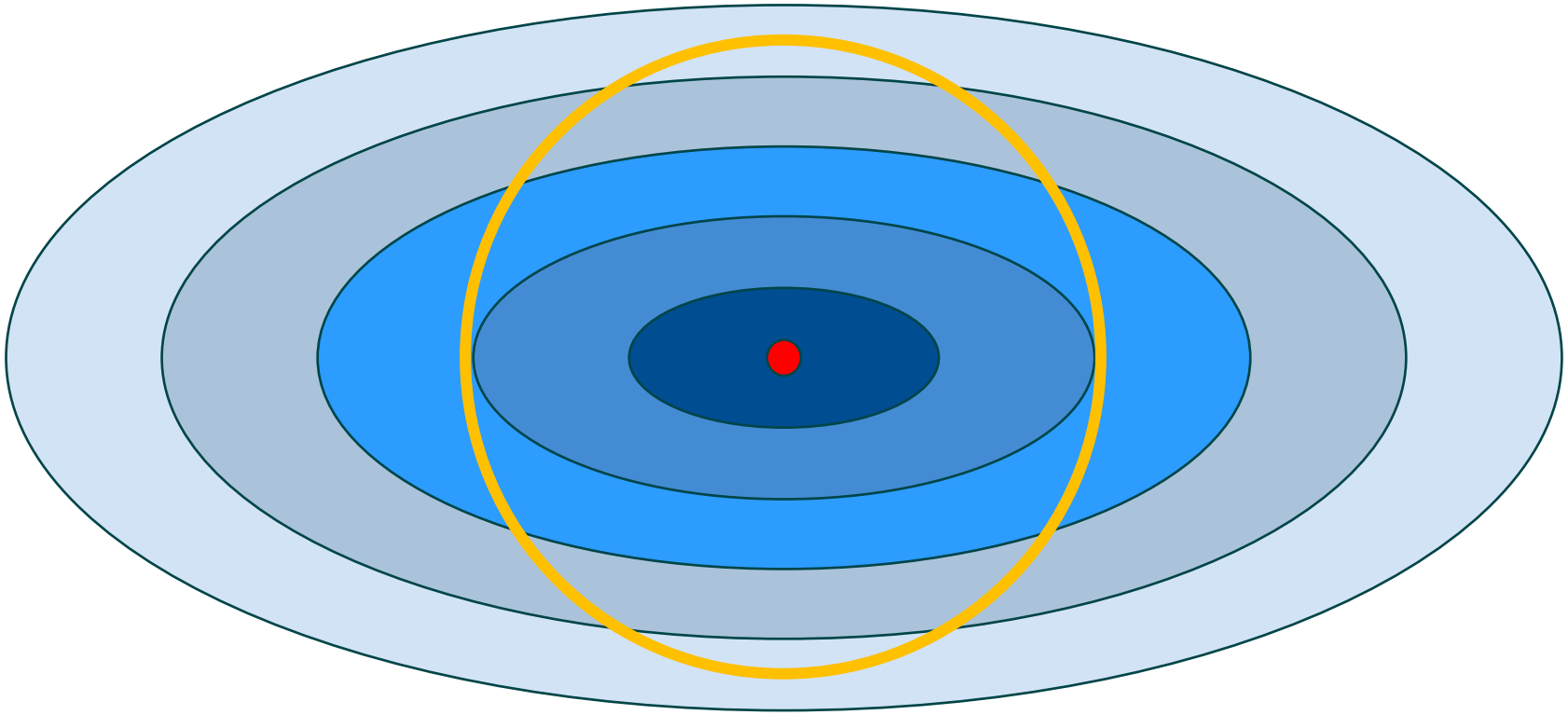
Mahalanobis Distance

How many σ are we away from the mean?



Mahalanobis Distance

How many σ are we away from the mean?



Distance of a point to a distribution

Recall **Variance**: $Var(X) = \sigma^2 = E[(X-\mu)^2] = \frac{\sum_{i=1}^{n}(x_i-\mu)^2}{n}$

Covariance generalizes notion of variance to dimensions > 1
What is the spread of the data in each dimension (and combination of dimensions)?

$$X = (x_1, \ldots, x_n), \qquad x_i \in \mathbb{R}^d$$

$$COV(x_{*j}, x_{*k}) = E[(\mathrm{x}_{*\mathrm{j}} - \mu_j)(\mathrm{x}_{*\mathrm{k}} - \mu_k)]$$

Covariance matrix: $\Sigma = E[(\mathrm{X} - \mu_X)(X - \mu_X)^T]$

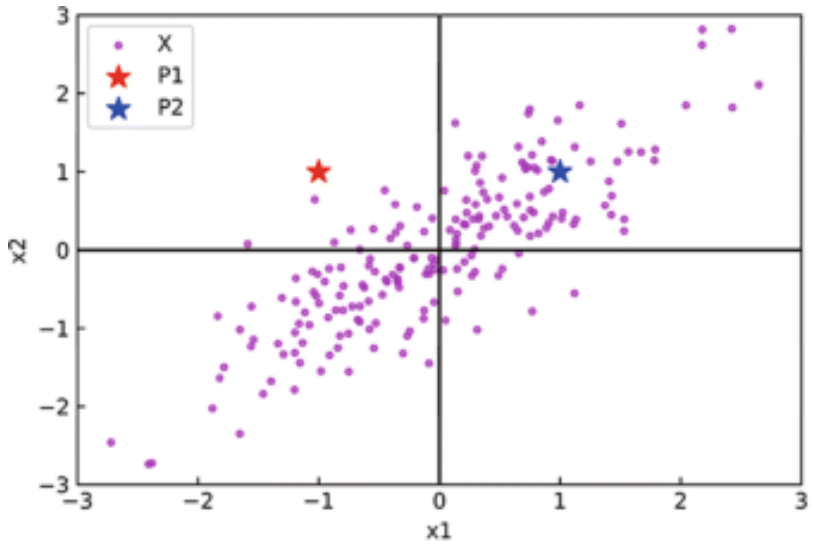
$$\Sigma = \begin{pmatrix} s_{11} & \cdots & s_{1d} \\ \vdots & \ddots & \vdots \\ s_{d1} & \cdots & s_{dd} \end{pmatrix}$$

$$s_{jk} = \frac{1}{n}\sum_{i=0}^{n}(x_{ij} - \mu_j)(x_{ik} - \mu_k)$$

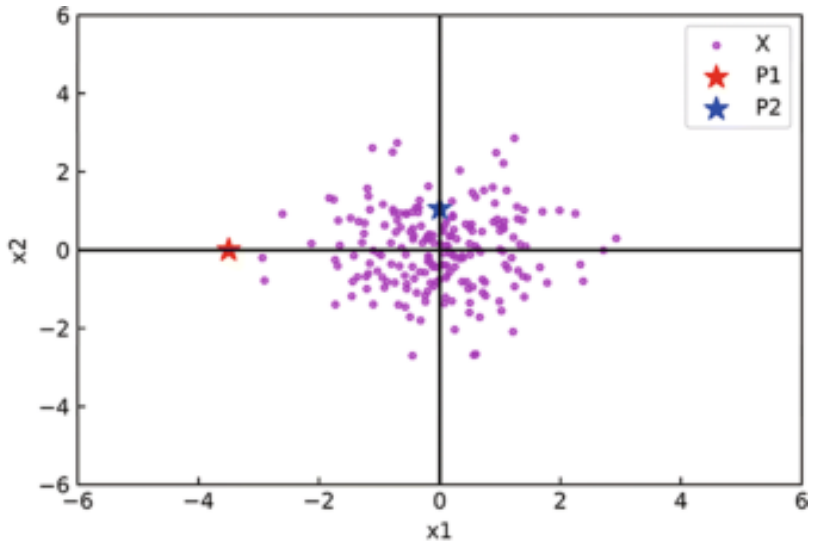
Mahalanobis Distance

Take covariance into account

$$D_M(x, \Sigma) = \sqrt{(x-\mu)^T \Sigma^{-1} (x-\mu)}$$



(a) Euclidean distance



(b) Mahalanobis distance

space mapped by $\Sigma^{-\frac{1}{2}}$

Visual Computing Institute | Dr. Isaak Lim
Computer Graphics and Multimedia
Data Analysis and Visualization

Visual Computing Institute
RWTH AACHEN UNIVERSITY

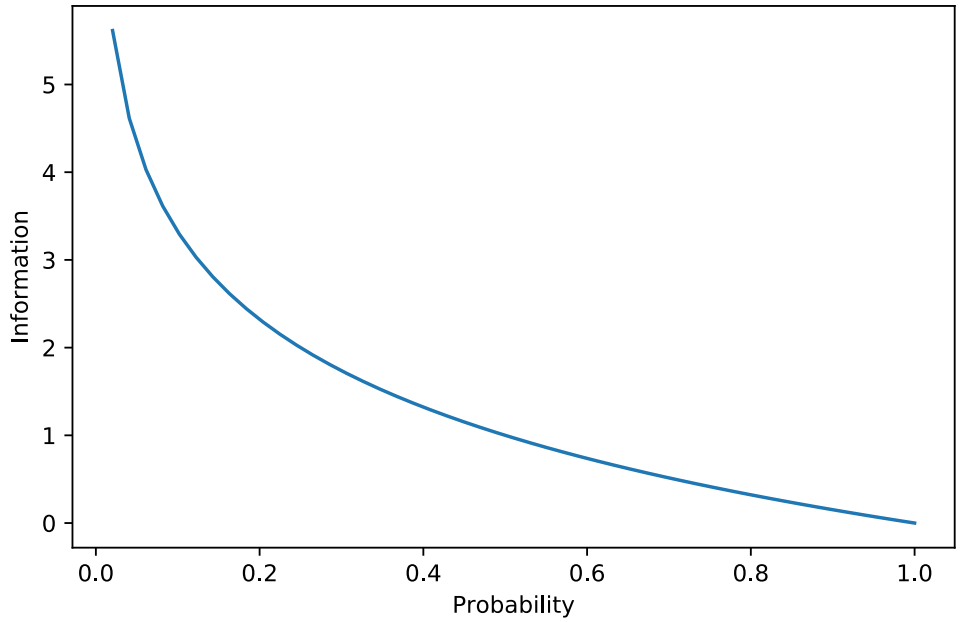
Comparing Distributions: Information Theoretic Measures

Somebody communicates to you an event:

- Low probability event: surprising / high information content
- High probability event: unsurprising / low information content

Information Theoretic Measures

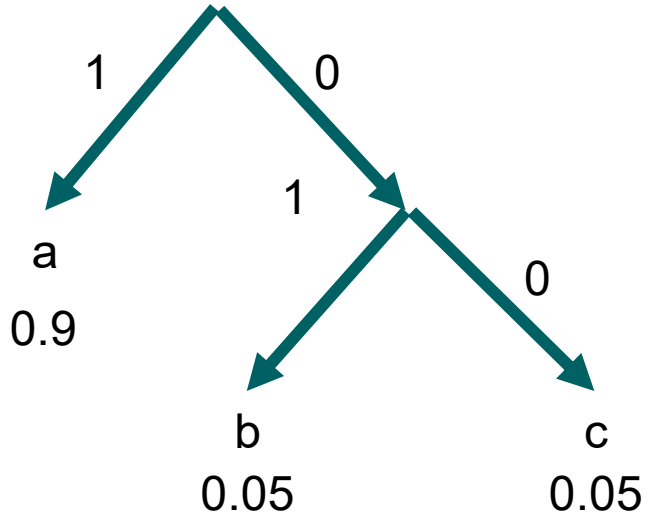
(self-)information: $I(x) = \log_2\left(\frac{1}{p(x)}\right) = -\log_2 p(x)$



Information Theoretic Measures

Entropy: expected information: $H(X) = E[I(X)] = -\sum_i p(x_i) \log_2 p(x_i)$

- Minimal number of bits needed to encode the information given an optimal encoding method
- Uniform distribution → maximal entropy



Measures between Distributions

Entropy: $H(p) = E_p[\log_2 \frac{1}{p}] = -\sum_{x \in X} p(x) \log_2 p(x)$

Cross-Entropy: $H(p,q) = E_p[\log_2 \frac{1}{q}] = -\sum_{x \in X} p(x) \log_2 q(x)$

- Expected message length when using approximated distribution q

Kullback-Leibler Divergence

Directed (asymmetric) measure between distributions that is widely used

Measures **relative entropy**:

$$D_{KL}(p\|q) = \sum_{x\in X} p(x)\log_2\left(\frac{p(x)}{q(x)}\right)$$

$$= \sum_{x\in X} p(x)(\log_2 p(x) - \log_2 q(x))$$

$$= \sum_{x\in X} p(x)\log_2 p(x) - \sum_{x\in X} p(x)\log_2 q(x) = H(p,q) - H(p)$$

non-negative
not a metric → can be used to make a metric (Jensen-Shannon distance)

Data Analysis: Outline

- Types of Data
- **Comparing Data**
  - feature spaces
  - similarity and dissimilarity measures
  - comparing histograms and distributions
- Dimensionality Reduction
- Clustering